

September 6, 2026 09:00 PM GMT

Zhongji InnoLight Co Ltd | Asia Pacific

Leading AI Enabler in Optics; Significant Growth Potential

We expect InnoLight to become the innovation leader in 2.4T/3.2T and NPO new products. Initiate H-share coverage at OW with price target of HK\$1,385, resume A-share coverage at OW with price target of Rmb1,205.

Structural AI optics upcycle: We expect a multi-year upcycle in AI data-center networking intensity as cloud service providers (CSPs) build “AI factories,” driving strong demand for high-end AI transceivers, 2026-2028. Thanks to the wider application of scale-out, scale-up and scale-across architectures, we expect accelerating adoption of AI transceivers, and we forecast demand to increase significantly, from ~20mn units in 2025 to ~73mn in 2026, ~141mn in 2027, and ~150mn in 2028, implying a 43% CAGR, 2026-2028.

Global leader in AI transceivers: InnoLight is a first-mover in 800G and 1.6T commercialization and appears likely to remain a pioneer in the next node (3.2T): The company’s long-term partnerships with global CSPs, close joint R&D, and differentiated new product introduction (NPI) support faster yield improvements and cost reductions, making InnoLight more competitive than peers on key metrics, such as time-to-market, time-to-volume and time-to-cost.

Potential new growth drivers beyond 2027: We expect accelerated new technology innovation, such as optical circuit switching (OCS) and near-packaged optics (NPO), to become technological breakthroughs in the coming years. Based on its industry leadership and technology expertise, we believe InnoLight is likely to capture these growth opportunities, and these new technologies could support the company’s growth beyond 2027. Success with NPO could also alleviate concerns about co-packaged optics (CPO) disruption.

Significant growth opportunity in 2026-28, driven by strong demand for 800G and 1.6T transceivers. InnoLight’s AI transceiver momentum is supported by industry growth and competitive advantages in time-to-market, time-to-volume and time-to-cost, underpinning superior top-line performance. From 2027 onward, 3.2T, NPO, and OCS could add new growth. Overall, we forecast 53% revenue and 64% earnings CAGRs, 2026–28.

Price targets: Our A-share target is derived from our residual income methodology. Our H-share target is based on an RMB/HKD FX rate of 1.15.

Key downside risks include 1) CPO disruption (timing and scope), 2) competitive intensity, and 3) geopolitics and tariff volatility.

MORGAN STANLEY ASIA LIMITED+

Andy Meng, CFA

Equity Analyst

Andy.Meng@morganstanley.com

+852 2239-7689

Betty Chen

Research Associate

Betty.H.Chen@morganstanley.com

+852 2239-7213

Zhongji InnoLight Co Ltd (300308.SZ, 300308 CS)

Greater China Technology Hardware | China

Stock Rating	Overweight
Industry View	In-Line
Price target	Rmb1,205.00
Up/downside to price target (%)	48
Shr price, close (Sep 3, 2026)	Rmb813.00
52-Week Range	Rmb1,416.88-337.00
Sh out, dil, curr (mn)	1,115
Mkt cap, curr (mn)	Rmb953,157
EV, curr (mn)	Rmb942,768
Avg daily trading value (mn)	Rmb25,651

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (Rmb)**	9.72	29.36	59.89	78.81
Prior EPS (Rmb)**	-	29.91	63.03	82.90
EPS (Rmb)\$	9.72	30.20	56.96	81.77
Revenue, net (Rmb mn)	38,240	107,904	196,339	251,279
EBITDA (Rmb mn)	14,244	45,017	86,963	113,301
ModelWare net inc (Rmb mn)	10,797	34,346	70,056	92,189
P/E	62.8	27.7	13.6	10.3
P/BV	22.8	8.8	5.5	3.7
RNOA (%)	73.7	192.9	178.8	142.4
ROE (%)	56.4	115.4	65.1	53.4
EV/EBITDA	46.9	20.7	10.3	7.3
Div yld (%)	0.2	0.2	0.5	1.1
FCF yld ratio (%)**	1.1	1.4	4.6	8.0
Leverage (EOP) (%)	(38.5)	(65.5)	(65.1)	(71.0)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework

\$ = Consensus data is provided by Refinitiv Estimates

** = Based on consensus methodology

e = Morgan Stanley Research estimates

Zhongji InnoLight Co Ltd (3308.HK)

Stock rating:	Overweight		
Price target:	HK\$1,385		
Shr price close (Sep 3, 2026):	HK\$995		
Upside:	39%		
Fiscal year ending	2026e	2027e	2028e
P/E:	29.47x	14.45x	10.98x

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Financial Summary

Exhibit 1: InnoLight: Financial summary

Financial Summary (Rmb mn)

Innolight (3308.HK)

Innolight (300308.SZ)

Income statements	2025	2026E	2027E	2028E
Net sales	38,240	107,904	196,339	251,279
COGS	(22,166)	(57,808)	(101,931)	(129,059)
Gross profit	16,074	50,096	94,409	122,220
Selling expenses	(226)	(489)	(785)	(1,005)
ADM expenses	(761)	(2,089)	(3,134)	(3,632)
R&D expenses	(1,615)	(3,249)	(4,583)	(5,528)
Operating Inc.	13,394	43,867	85,513	111,551
Finance cost	(183)	(1,362)	200	200
Pre-tax Income	13,600	43,224	85,713	111,751
Income Tax	(2,020)	(6,396)	(12,857)	(16,763)
Net income	10,797	34,346	70,056	92,189
Modelware EPS (RMB)	9.72	29.36	59.89	78.81
EPS (RMB)	9.72	29.36	59.89	78.81
EBITDA	14,244	45,017	86,963	113,301
Balance sheet	2025	2026E	2027E	2028E
Cash	10,987	69,994	111,953	180,503
Trade receivables	6,350	17,917	32,601	41,723
Inventories	12,681	33,071	58,314	73,834
Other current assets	1,067	1,067	1,067	1,067
Current Assets	31,084	122,049	203,935	297,127
Property, plant and equipment	8,503	10,353	11,903	13,153
Intangible assets	2,316	2,316	2,316	2,316
Other assets	3,386	3,386	3,386	3,386
Total Assets	45,289	138,104	221,540	315,983
Trade payables	7,800	20,343	35,871	45,418
Other payables	2,672	2,672	2,672	2,672
Bank loans - due within one year	1,086	1,086	1,086	1,086
Current Liabilities	11,558	24,101	39,629	49,176
Bank loans - due after one year	510	510	510	510
Other liabilities	1,599	1,599	1,599	1,599
Total Liabilities	13,668	26,211	41,738	51,285
Share capital	1,111	46,111	46,111	46,111
Reserves and minority interests	28,654	61,444	126,552	208,648
Shareholders' equity	29,765	107,555	172,663	254,759
Total Liab./Shrhldr's Equity	45,289	138,104	221,540	315,983
Cashflow statements	2025	2026E	2027E	2028E
Cashflow from operations	10,896	17,201	50,107	81,843
Profits from operation	13,600	43,224	85,713	111,751
Depreciation & Amortization	850	1,150	1,450	1,750
Change in Working Capital	(2,029)	(19,415)	(24,399)	(15,095)
Others	(1,524)	(7,758)	(12,657)	(16,563)
Cashflow from investing	(2,619)	(3,000)	(3,000)	(3,000)
(Purchases) sale of PPE	(2,752)	(3,000)	(3,000)	(3,000)
Others	133	-	-	-
Cashflow from financing	(2,144)	44,807	(5,148)	(10,293)
Increase in bank loans	(1,114)	-	-	-
Dividends & Interests	(1,083)	(1,556)	(4,948)	(10,093)
Others	(2,113)	43,251	(10,096)	(20,386)
Net change in cash	5,956	59,008	41,959	68,550

Ratio analysis	2025	2026E	2027E	2028E
YoY growth %				
Sales	60%	182%	82%	28%
Operating profits	127%	228%	95%	30%
Pretax profits	125%	218%	98%	30%
EBITDA	117%	216%	93%	30%
Net profits	109%	218%	104%	32%
Margins %				
Gross margin	42.0%	46.4%	48.1%	48.6%
Operating margin	35.0%	40.7%	43.6%	44.4%
Pretax margin	35.6%	40.1%	43.7%	44.5%
EBITDA margin	37.2%	41.7%	44.3%	45.1%
Net margin	28.2%	31.8%	35.7%	36.7%
Return %				
ROAE	44%	50%	50%	43%
ROAA	29%	37%	39%	34%
Gearing %				
Net Debt/Equity	-32%	-64%	-64%	-70%
Liabilities/Equity	46%	24%	24%	20%
Liabilities/Assets	30%	19%	19%	16%
WC ratio %				
AR/NR Turnover (days)	53	41	47	54
AP/NP Turnover (days)	93	89	101	115
Inventory Days	162	144	164	187
Cash conversion cycle (days)	122	97	110	126

Segment analysis	2025	2026E	2027E	2028E
Revenue				
200G and Below	435	391	352	317
400G	12,077	8,547	5,424	4,118
800G	18,922	40,815	48,879	45,914
1.6T	6,328	57,624	128,105	125,581
Access	479	527	580	637
Revenue mix				
200G and Below	1%	0%	0%	0%
400G	32%	8%	3%	2%
800G	49%	38%	25%	18%
1.6T	17%	53%	65%	50%
Access	1%	0%	0%	0%
Valuation	2025	2026E	2027E	2028E
P/E	88.1	27.7	13.6	10.3
P/B	31.9	8.8	5.5	3.7
P/S	24.9	8.8	4.8	3.8
EV/EBITDA	66.1	20.9	10.8	8.3

Source: Company data, Morgan Stanley Research (E) estimates.

Risk Reward – Zhongji Innolight Co Ltd (300308.SZ)

Leading AI Enabler in Optics; Significant Growth Potential

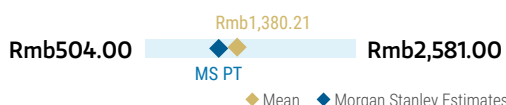
PRICE TARGET Rmb1,205.00

Residual income model, probability-weighted 30% bull, 50% base, 20% bear - skew reflects expected timeline for large-scale CPO application.

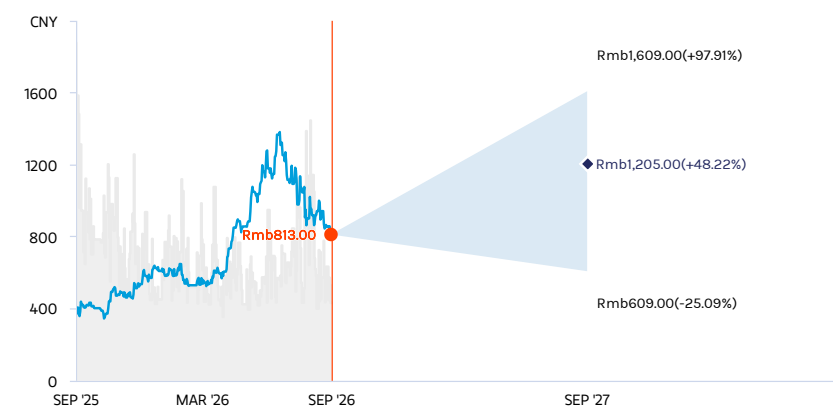
Key assumptions:

- CoE and terminal growth rate: 10% and 3.5% across scenarios
- High-end transceivers demand CAGR in 2026-2028: bull 46%, base 43%, bear 43%
- 2028-36e net profit CAGR: bull 12%, base 8%, bear 1%.

Consensus Price Target Distribution



RISK REWARD CHART



Source: Refinitiv, Morgan Stanley Research

BULL CASE

Rmb1,609.00

26x bull case 2027e EPS

Stronger 800G and 1.6T ramp-up and share gains within new clients: We assume stronger demand for 800G and 1.6T, which would be key catalysts for the company in 2025-2026. We expect Innolight to enjoy faster-than-expected growth together with market share gains. CPO breakthrough to be further delayed while NPO technology becomes mature ahead of that.

BASE CASE

Rmb1,201.00

20x base case 2027e EPS

Robust earnings growth in 2026-2027: We maintain our positive demand outlook for 800G and 1.6T transceivers. We expect 800G and 1.6T demand to drive robust earnings growth in 2026-2027. CPO breakthrough will arrive in 2027 or later.

BEAR CASE

Rmb609.00

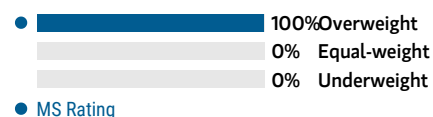
12x bear case 2027e EPS

Intense competition and weaker-than-expected orders: We assume slower demand for 800G and 1.6T products. Innolight also faces intense competition, resulting in lower market share, margins, and ASPs. In this case, it would also de-rate. If there is any order cut, the stock's valuation could fall even further. CPO to achieve earlier-than-expected breakthrough, as a result, the company's revenue growth grows more slowly.

OVERWEIGHT THESIS

- We expect a multi-year upcycle in AI data-center networking intensity as cloud service providers (CSPs) build "AI factories", driving strong demand for high-end AI transceivers, 2026-2028.
- Innolight is a first mover in 800G and 1.6T commercialization and is likely to remain a pioneer in the next node (3.2T).
- We expect accelerated new technology innovation, such as optical circuit switching (OCS) and near-packaged optics (NPO), to become technological breakthroughs beyond 2027.
- On the basis of our probability-weighted bull-base-bear valuation and bull/bear/base case scenario values, we see upside in the stock.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: *Positive*

View descriptions of Risk Rewards Themes [here](#)

Risk Reward – Zhongji Innolight Co Ltd (300308.SZ)

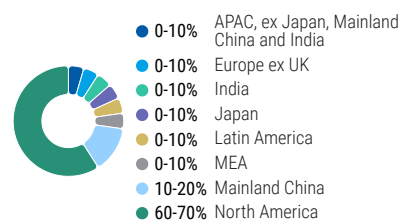
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
800G Revenue YoY (%)	127.9	115.7	19.8	(6.1)
1.6T Revenue YoY (%)	0.0	810.7	122.3	(2.0)
200G and Below Revenue YoY (%)	(69.8)	(10.0)	(10.0)	(10.0)
400G Revenue YoY (%)	(11.5)	(29.2)	(36.5)	(24.1)

INVESTMENT DRIVERS

- 200G, 400G, 800G, and 1.6T transceiver demand in the datacom market
- Potential cost control through internal operational management
- Potential new client breakthrough in auto LiDAR market
- Developments in Co-Packaged Optics

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

3/5
MOST 3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Later breakthrough in CPO
- Success with NPO
- Stronger-than-expected cloud infrastructure demand
- Share gains from new customers
- Better-than-expected cost control

RISKS TO DOWNSIDE

- Earlier breakthrough in CPO or other new technology
- ASP declining faster than cost controls, leading to margin pressure
- Geopolitical issues that cause supply chain disruption
- Competitive intensity

OWNERSHIP POSITIONING

Inst. Owners, % Active **88%**

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (Rmb, mn) 84,065.0 **107,903.9** 124,239.9
104,259.9

EBITDA (Rmb, mn) 37,553.6 **45,017** 51,018.9
44,011.5

Net income (Rmb, mn) 27,676.0 **34,346** 40,531.0
33,834.4

EPS (Rmb) 23.1 **29.4** 36.5
30.2

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Zhongji Innolight Co Ltd (3308.HK)

Leading AI Enabler in Optics; Significant Growth Potential

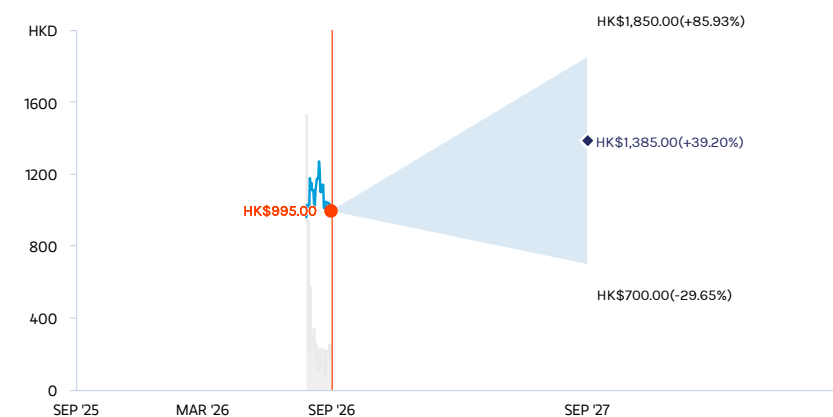
PRICE TARGET HK\$1,385.00

Residual income model, probability-weighted 30% bull, 50% base, 20% bear - skew reflects expected timeline for large-scale CPO application. We apply an RMB/HKD FX rate of 1.15 to arrive at our H-share price target.

Key assumptions:

- CoE and terminal growth rate: 10% and 3.5% across scenarios
- High-end transceivers demand CAGR in 2026-2028: bull 46%, base 43%, bear 43%
- 2028-36e net profit CAGR: bull 12%, base 8%, bear 1%.

RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect a multi-year upcycle in AI data-center networking intensity as cloud service providers (CSPs) build "AI factories", driving strong demand for high-end AI transceivers, 2026-2028.
- InnoLight is a first mover in 800G and 1.6T commercialization and is likely to remain a pioneer in the next node (3.2T).
- We expect accelerated new technology innovation, such as optical circuit switching (OCS) and near-packaged optics (NPO), to become technological breakthroughs beyond 2027.
- On the basis of our probability-weighted bull-base-bear valuation and bull/bear/base case scenario values, we see upside in the stock.

Risk Reward Themes

New Data Era: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

HK\$1,850.00

26x bull case 2027e EPS

Stronger 800G and 1.6T ramp-up and share gains within new clients: We assume stronger demand for 800G and 1.6T, which would be key catalysts for the company in 2025-2026. We expect Innolight to enjoy faster-than-expected growth together with market share gains. CPO breakthrough to be further delayed while NPO technology becomes mature ahead of that.

BASE CASE

HK\$1,380.00

20x base case 2027e EPS

Robust earnings growth in 2026-2027: We maintain our positive demand outlook for 800G and 1.6T transceivers. We expect 800G and 1.6T demand to drive robust earnings growth in 2026-2027. CPO breakthrough will arrive in 2027 or later.

BEAR CASE

HK\$700.00

12x bear case 2027e EPS

Intense competition and weaker-than-expected orders: We assume slower demand for 800G and 1.6T products. Innolight also faces intense competition, resulting in lower market share, margins, and ASPs. In this case, it would also de-rate. If there is any order cut, the stock's valuation could fall even further. CPO to achieve earlier-than-expected breakthrough, as a result, the company's revenue growth grows more slowly.

Risk Reward – Zhongji Innolight Co Ltd (3308.HK)

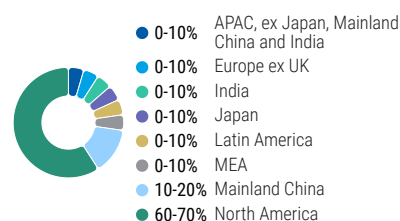
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
800G Revenue YoY (%)	127.9	115.7	19.8	(6.1)
1.6T Revenue YoY (%)	0.0	810.7	122.3	(2.0)
200G and Below Revenue YoY (%)	(69.8)	(10.0)	(10.0)	(10.0)
400G Revenue YoY (%)	(11.5)	(29.2)	(36.5)	(24.1)

INVESTMENT DRIVERS

- 200G, 400G, 800G, and 1.6T transceiver demand in the datacom market
- Potential cost control through internal operational management
- Potential new client breakthrough in auto LiDAR market
- Developments in Co-Packaged Optics

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

RISKS TO PT/RATING

RISKS TO UPSIDE

- Later breakthrough in CPO
- Success with NPO
- Stronger-than-expected cloud infrastructure demand
- Share gains from new customers
- Better-than-expected cost control

RISKS TO DOWNSIDE

- Earlier breakthrough in CPO or other new technology
- ASP declining faster than cost controls, leading to margin pressure
- Geopolitical issues that cause supply chain disruption
- Competitive intensity

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (Rmb, mn) ◆ 107,903.9

Note: There are not sufficient brokers supplying consensus data for this metric

EBITDA (Rmb, mn) ◆ 45,017

Note: There are not sufficient brokers supplying consensus data for this metric

Net income (Rmb, mn) ◆ 34,346

Note: There are not sufficient brokers supplying consensus data for this metric

EPS (Rmb) ◆ 29.4

Note: There are not sufficient brokers supplying consensus data for this metric

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Executive Summary

InnoLight is a global leader in AI optical transceivers, with first-mover execution across 800G and 1.6T, deep partnerships with top-tier cloud service providers (CSPs), strong R&D capabilities and efficient execution that helps it outperform in new products' time-to-market, time-to-volume and time-to-cost. We expect the company to benefit from a multi-year upcycle supported by scale-out, scale-up, and scale-across applications – aimed at making AI factories available for everyday AI computing.

We set our H-share price target at HK\$1,385 and A-share price target at Rmb1,205, derived from residual income valuation. Compared with industry peers, InnoLight's H-share is trading at 14.4x 2027ex P/E – a 58% discount vs. global peers. InnoLight's A-share is trading at 13.6x 2027e P/E – a 75% discount to A-share industry peers. We believe such valuation discount reflects investor concerns that component shortages and CPO disruption could pose key risks to rapid revenue and earnings growth.

Key Takeaways

Structural demand tailwinds: We expect AI data-center networking upgrades (scale-out in 2026, scale-up from 2027, scale-across ramping over time) to drive a rapid increase in 800G/1.6T industry volumes (from ~20mn units in 2025 to ~73mn in 2026 and ~141mn in 2027).

Market leadership: First-mover in 800G/1.6T development and commercialization, with potential to remain a pioneer in 3.2T; leading share in the most advanced AI optics (e.g., ~40% in 1.6T, we estimate). Potential pioneer of the next-generation product (3.2T).

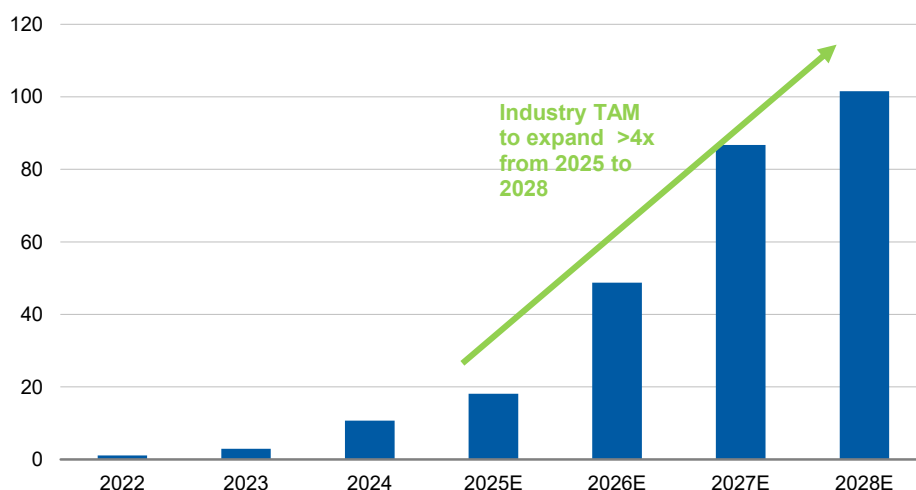
Customer and execution edge: Long-term partnerships with top-tier CSPs and AI computing solution providers, enabling early access to requirements and joint R&D; differentiated NPI/early manufacturing engagement to accelerate learning cycles.

R&D breadth and depth: InnoLight's capabilities span key optical technology areas (e.g., silicon photonics, coherent optics, XPO, linear pluggable optics (LPO) and linear receive optics (LRO), and NPO- and CPO-related work). These capabilities are supported by a global talent base.

Faster industrialization through differentiated NPI: Early-stage manufacturing teams work closely with R&D teams to shorten feedback loops and improve yields and costs faster than peers.

Scaled manufacturing and resilient supply chain: InnoLight's large production capacity supports large orders and procurement advantages. Its ongoing overseas capacity expansion and multi-region sourcing are intended to improve resilience.

Potential new growth opportunities beyond 2027: OCS and NPO could add incremental demand and/or reduce disruption risk versus a pure pluggable-module roadmap, offering upside beyond our base-case assumptions.

Exhibit 2: Global AI transceiver TAM to expand >4x, 2025-2028

Source: Company data, Omdia, Morgan Stanley Research (E) estimates. Note: Units in US\$ bn.

Industry Demand Outlook (2026–2028): AI Factory Networking Drives High-End Optics

We expect AI infrastructure build-outs to materially increase optical connectivity intensity and accelerate adoption of 800G and 1.6T transceivers. In our base case scenario, as described in this report, high-end transceiver demand rises from ~20mn units in 2025 to ~73mn in 2026, ~141mn in 2027, and ~150mn in 2028, driven by three architectural shifts:

Scale-out (rack-to-rack connectivity within a data center; we expect it to be broadly adopted from 2026): Scale-out expands multi-tier interconnects across racks and switch banks. It increases connection points (top-of-rack (ToR)-to-switch, switch-to-switch, and GPU rack-to-switch links), pushing both unit volumes and speed mix toward 800G/1.6T as GPU cluster size and switch ASIC bandwidth rise.

Scale-up (inside a rack or among tightly coupled accelerators; we expect it to be broadly adopted from 2027): Scale-up addresses the highest traffic share (noted as ~80–90% of in-rack/XPU-to-XPU communication). As copper reaches density and bandwidth limits, optical adoption becomes more likely, creating a new demand pool and raising performance requirements for latency, power, and cost.

Scale-across (among geographically distributed data centers): Scale-across helps CSPs overcome power and space constraints and supports distributed training and inference. Rising east-west data movement increases demand for long-reach, high-speed optical links and expands 800G/1.6T use cases beyond a single facility.

On June 1, 2026, Alphabet Inc. announced an equity offering totaling US\$80 billion in expected aggregate amount, primarily to fund investments in AI compute infrastructure. Due to higher-than-expected investor demand, the offering was subsequently upsized and priced at US\$84.75 billion on June 2, 2026. This marks Alphabet's first equity financing in 21 years. Alphabet intends to use the funds to support the expansion of data centers, computing capacity, and systems for training and deploying LLM. We believe such recent developments may help strengthen the AI infrastructure capex outlook, which could further boost AI transceiver demand.

Potential new growth opportunities: OCS and NPO

Optical Circuit Switching (OCS): Emerging deployments in AI data centers (e.g., greenfield spine replacement, TPU interconnects, and potential scale-up applications inside racks) can create incremental transceiver attachment points. OCS is generally complementary to optics modules and may also raise module requirements (e.g., span budgets, insertion loss tolerance, and power levels).

Near-Packaged Optics (NPO): Moves the optical engine close to the switch ASIC on the same board, improving signal integrity and power efficiency versus traditional pluggable modules, while avoiding some of the manufacturing complexity of full CPO. A successful NPO path could help incumbent transceiver suppliers remain relevant if the industry shifts toward higher integration.

Growth Outlook

We expect InnoLight's revenue and earnings to grow strongly in 2026–2028, underpinned by: (1) rapid end-demand expansion for high-end optics; and (2) company-specific execution advantages (in time-to-market, time-to-volume, and time-to-cost). In the AI transceiver business, 800G and 1.6T products are the near-term growth engines – supporting growth in 2026, with 1.6T increasingly becoming the key driver as AI infrastructure upgrades accelerate into 2027.

Beyond the current pluggable-module cycle, NPO, OCS and other next-generation optical architectures could add incremental growth from 2027 onward.

Our model assumptions imply InnoLight will deliver 53% revenue and 64% earnings CAGRs, 2026–2028.

Key Risks and Watch Items

CPO disruption risk (timing and magnitude): CPO is a structural architectural shift that could displace pluggable modules and reshape the ecosystem. However, this report highlights meaningful hurdles – yield and thermal challenges, a large cost premium versus pluggables, ecosystem immaturity, and a larger “failure domain” that complicates operations – suggesting large-scale adoption is more likely in 2027–2028 and beyond, with outcomes best framed via scenario analysis.

Competitive intensity and customer diversification: As 800G/1.6T become mainstream, more vendors are likely to qualify and expand capacity. This could pressure market share and pricing, even if absolute volumes continue to rise.

Geopolitics, tariffs, and supply chain volatility: High overseas revenue exposure and reliance on global component sourcing (including constrained components such as electro-absorption modulated lasers (EML)) create sensitivity to policy changes, trade frictions, and broader supply chain restructuring. The recent discussion of a potential FCC ban on imports of Chinese optical transceivers into the US is a current example (see: [Potential geopolitical tensions is a macro risk factor](#)).

Investment Positives

Global leader in the AI transceiver industry

- We view InnoLight as an established global leader in the AI transceiver industry. The company has built a strong track record over the past decade, including:
- 1) InnoLight was an early mover in developing 800G and 1.6T transceivers and was also the first company globally to achieve successful commercial production. We believe that InnoLight's expertise positions it to be a pioneer in the upcoming 3.2T transceiver product cycle.
 - 2) InnoLight has established long-term partnerships with all the global top-tier CSPs and AI computing solutions providers. These partnerships help ensure that InnoLight receives advanced new product requirements early and can conduct joint R&D well ahead of most other industry peers.
 - 3) InnoLight maintains leading R&D capabilities, supported by a world class global talent base. The company has established leading R&D capabilities across key technology fields, including i) SiPh, ii) coherent optics, iii) XPO, iv) LPO/LRO, and v) NPO/CPO.
 - 4) InnoLight has established a differentiated NPI team setup to accelerate new product innovation. Unlike most other peers, the company has set up early-stage manufacturing teams that work closely with R&D teams to collect real-time feedback and improve new products more efficiently.
 - 5) InnoLight also maintains production scale and supply capacity at the forefront of the industry. Its scale advantages significantly enhance its ability to undertake large orders while reducing manufacturing and procurement costs. This supports the company's ability to maintain market leadership.
 - 6) InnoLight has expanded overseas capacity aggressively, positioning it to capture global growth opportunities. In 2025, 91% of the company's revenue was from overseas markets. The company has also established a flexible and resilient supply chain, which helps it source high-quality components from suppliers around the world.

Exhibit 3: InnoLight: Global leader in the AI transceiver industry



Source: Company data. Note: (1) According to CIC; (2) By revenue; (3) As of March 31, 2026.

Given these competitive advantages, we believe that InnoLight is well positioned to outperform most industry peers across the three key metrics that customers care about most: 1) time-to-market; 2) time-to-volume; and 3) time-to-cost. This also explains why InnoLight has become a fast-rising leader in the transceiver industry. As illustrated in [Exhibit 4](#), InnoLight ranked outside the global top 10 in 2010 but rose quickly to No. 2 in 2018, only ten years after its debut. In 2023, the company became the world's No. 1 transceiver supplier and has since maintained that ranking.

Exhibit 4: InnoLight: Global leader in the AI transceiver industry

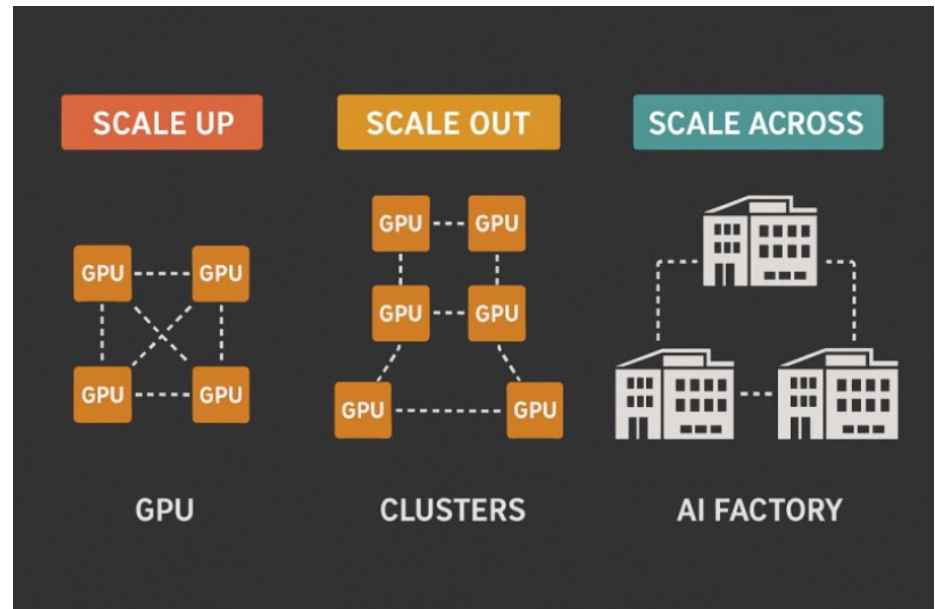
Ranking of Top 10 Transceiver Suppliers				
Rank	2010	2018	2023	2024
1	Finisar	Finisar	Innolight	Innolight
2	Opnext	Innolight	Coherent	Coherent
3	Sumitomo	Hisense	Huawei	Eoptolink
4	Avago	Accelink	Cisco	Huawei
5	Source Photonics	FOIT (Avago)	Accelink	Cisco
6	Fujitsu	Lumentum/Oclaro	Hisense	Accelink
7	JDSU	Acacia	Eoptolink	Hisense
8	Emcore	Intel	HG Genuine	Marvell
9	WTD	AOI	Source Photonics	HG Genuine
10	NeoPhotonics	Sumitomo	Marvell	Source Photonics

Source: : LightCounting, Morgan Stanley Research.

Because the rankings above also include non-AI transceiver shipments, such as telecom demand, we believe InnoLight's market share in the AI transceiver industry could be even higher. For the most advanced AI transceivers, such as 1.6T, we estimate the company's market share could be near ~40%. Thus, we believe InnoLight could maintain a leading position in the upcoming 3.2T product cycle.

Significant growth opportunity from AI factory build-outs, driven by scale-up, scale-out, and scale-across

We believe scale-up and scale-out technologies will be widely applied in data center architectures, significantly improving overall data processing capacity and efficiency. As a result, demand for transceivers is also likely to increase materially. We forecast demand for high-end (800G and 1.6T) transceivers to increase from 20mn units in 2025 to 73mn in 2026, 141mn in 2027, and 150mn in 2028. This rapid volume increase is the key fundamental driver of revenue and earnings growth for the industry.

Exhibit 5: Scale-up, Scale-out, and Scale-Across

Source: Morgan Stanley Research.

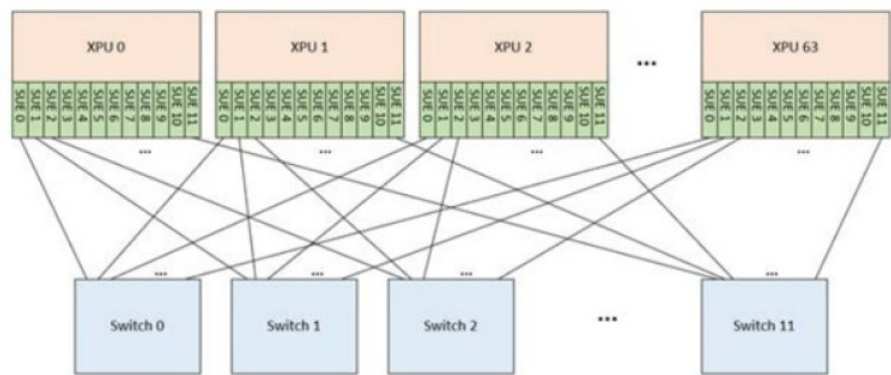
1) Scale-Out Architecture (Rack-to-Rack)

Scale-out networks are currently deployed in AI data centers to connect racks within a data center, enabling efficient scaling for training across hundreds of thousands of GPUs. We expect widespread scale-out adoption to begin in 2026 (see: [AI Transceivers: Growth Dominates Disruption](#)). Scale-out drives transceiver demand through i) network architecture expansion requiring extensive rack-to-rack connectivity, ii) growing data center clusters with bigger Gen AI clusters, iii) increasing bandwidth requirements pushing the market toward 800G and 1.6T products, and iv) multiple connection points at various levels of data center infrastructure.

2) Scale-Up Architecture (In-Rack GPU-to-GPU)

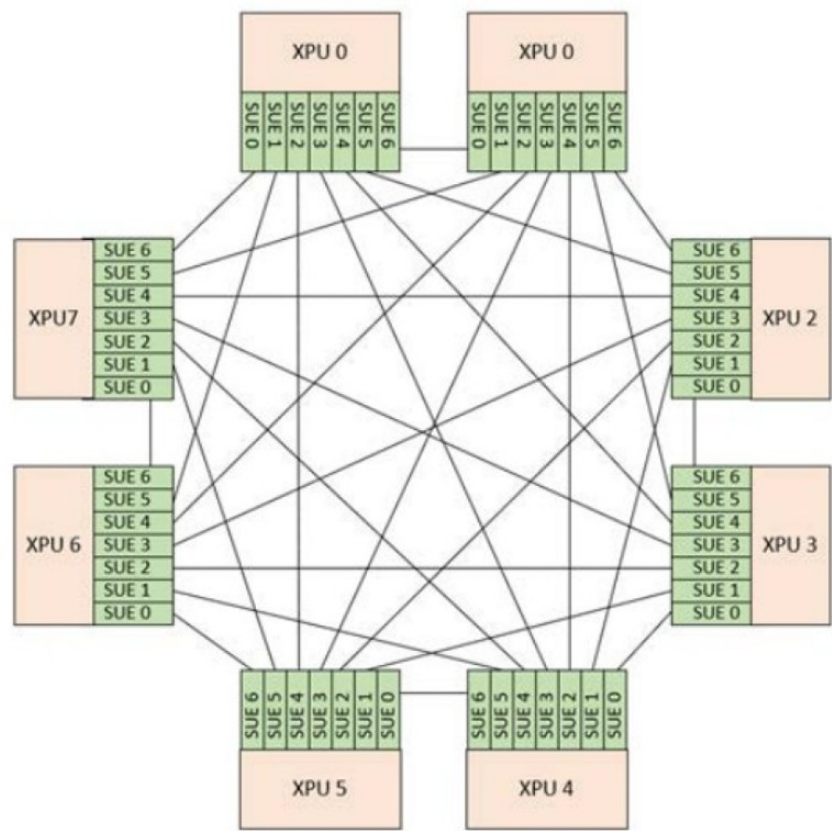
Scale-up networking refers to high-speed, low-latency interconnects that link GPUs or other accelerators within a single server or tightly coupled rack. Currently, this traffic is primarily handled by passive copper, but the transition to optical solutions for scale-up applications creates new opportunities for transceiver suppliers. [We expect scale-up to be widely applied from 2027](#). Network saturation is a critical issue, with up to 33% of GPU time wasted waiting for network availability, translating to US\$10,000+ per GPU per year in idle costs and driving urgent demand for higher-capacity transceivers.

Exhibit 6: Example Single Hop Switched SUE Deployment

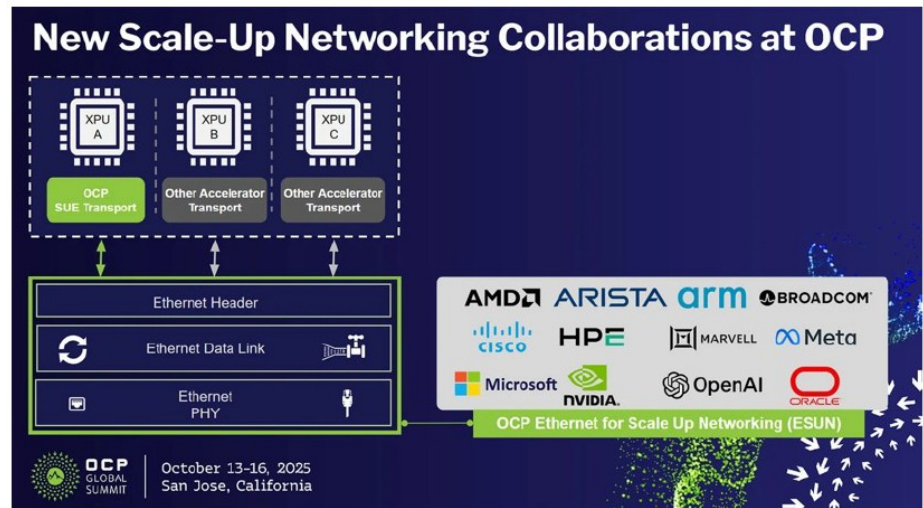


Source: Scale Up Ethernet Framework White Paper.

Exhibit 7: Example of Mesh SUE Deployment



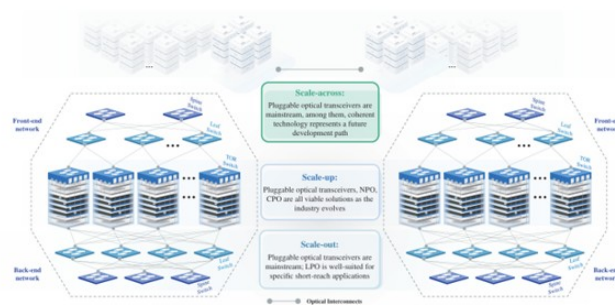
Source: Scale Up Ethernet Framework White Paper.

Exhibit 8: ESUN Announcement Partners

Source: OCP.

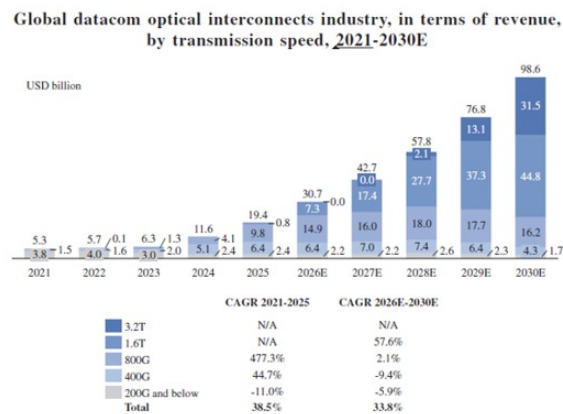
3) Scale-Across Architecture (Data Center-to-Data Center)

Scale-across refers to connecting geographically distributed data centers to operate as a single logical system. The amount of data leaving a data center is growing from approximately 5% of traffic to closer to 20% now, under the inference working scenario, creating substantial upward pressure on demand estimates. Scale-across networks depend heavily on optical fiber and optical transceivers for efficient cross-data center collaboration, particularly for long-distance links. Scale-across networks require substantially higher bandwidth than traditional WAN connections – scale-out networks have approximately 10x more capacity than WAN, and scale-up networks have another 10x beyond that.

Exhibit 9: Three AI deployment scenarios within data center architectures

Source: China Insights Consultancy Limited.

Exhibit 10: 800G to drive material growth in 2024-26, with 1.6T becoming another key growth driver in 2026-2028



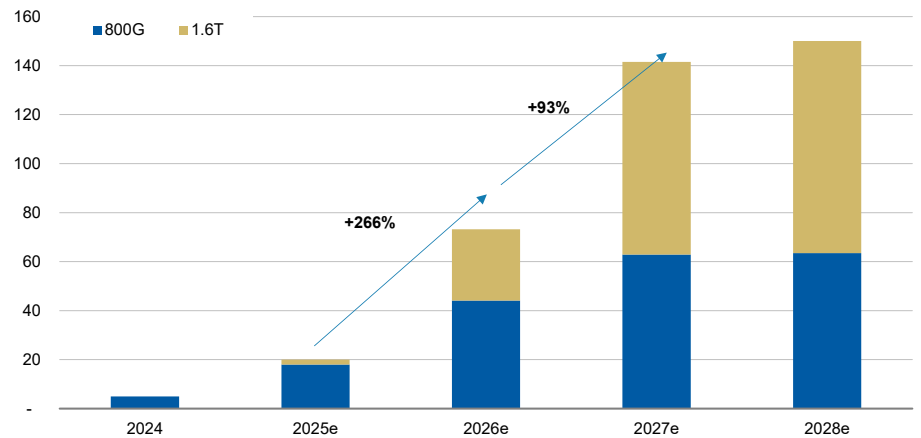
Source: LightCounting (E) estimates, China Insights Consultancy Limited. Note: Historical and forecast figures are broadly based on data from LightCounting, while the forecast breakdown is based on CIC estimates.

Bullish transceiver demand outlook in 2026-2028

By 2025, 800G products were fully commercialized and widely adopted, and 1.6T products achieved mass production. With GPU delivery delays easing in 2025, 1.6T transceivers began scale production in 2H25. More transceiver companies also completed R&D for 1.6T products and began new product verification in 2025. If this continues, we believe more 1.6T manufacturers will be added to qualified vendor lists, supporting strong 1.6T volume growth that could exceed LightCounting's forecast.

We expect scale-out to be widely applied from 2026, scale-up from 2027, and scale-across to expand from small-scale to large-scale deployment over the next several years. Based on this view, we forecast demand for high-end (800G and 1.6T) transceivers to increase from 20mn units in 2025 to 73mn units in 2026, up 266% YoY, sustaining the strong growth momentum observed in 2025. After that, we expect high-end transceiver demand to increase further, to 141mn units in 2027, implying 93% YoY growth.

Exhibit 11: Bullish transceiver demand outlook, 2026-2028



Source: Company data, Morgan Stanley Research (E) estimates.

OCS and NPO could become additional drivers beyond existing pluggable transceivers

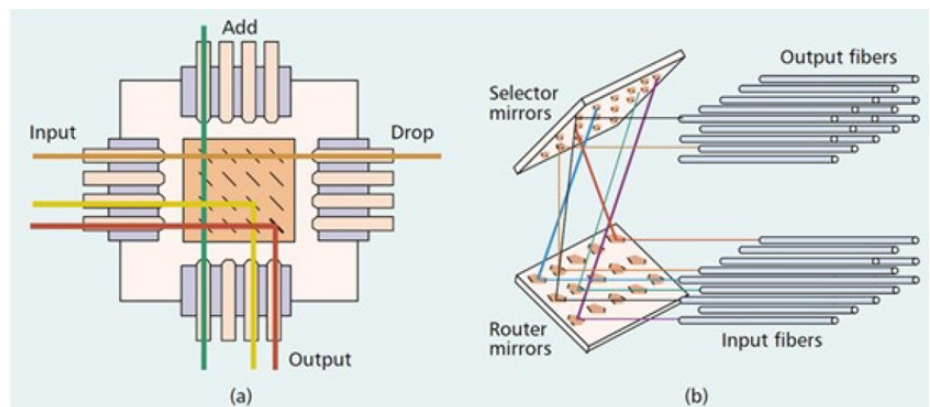
We believe the optical transceiver industry has entered a new innovation cycle, with several new technologies emerging amid surging AI infrastructure demand – including CPO, scale-out, scale-up and scale-across. There has been meaningful R&D progress in recent years, and new applications are likely to be introduced in the coming years.

In addition, we expect continued technology innovation to drive industry demand. We believe OCS and NPO are likely to become two additional technological breakthroughs in the coming years. We have not reflected any demand/volume impact in our base scenario; however, if these two technologies break through earlier than expected, we would expect a stronger positive impact on the transceiver industry, supporting the industry bull case scenario.

OCS – another emerging positive driver for transceiver demand

Optical circuit switching (OCS) is an emerging networking technology that routes optical signals directly without converting them to electrical signals, eliminating the need for traditional optical-electrical-optical (OEO) conversion. This technology is gaining significant traction in AI data centers and high-performance computing environments because it can deliver ultra-low latency, reduced power consumption, and high reliability.

Exhibit 12: OCS Example with MEMS Mirrors Forming CircuitSwitches



Source: researchgate.net.

OCS is a new network technology, and several technology roadmaps are currently under R&D, including:

- **MEMS-based switches:** Lumentum and others use micro-electro-mechanical systems (MEMS) technology, which offers precision, reliability, and cost-effectiveness. Lumentum's MEMS technology has more than one trillion field mirror operating hours in telecom applications.
- **Liquid crystal technology:** Coherent Corp. uses non-mechanical, field-proven digital liquid-crystal technology that has been deployed for many years in telecom applications.
- **Silicon photonics:** iPrionics offers programmable photonics approaches for OCS applications.

Exhibit 13: Major OCS Solutions in the R&D Pipeline

Vendor	Product	Port Configuration	Technology	Status
Lumentum (LITE)	R300 OCS	300x300	MEMS	Strong market reception
Lumentum (LITE)	R64 OCS	64x64 (Scalable to 72x72)	MEMS	Sampling Q4 2025, GA H2 2026
Coherent (COHR)	OCS Platform	64x64 and 320x320	Liquid Crystal	Shipped to 7 customers as of Nov 2025
FS	DCS - W Series	8x8, 16x16, 32x32	MEMS	Launched Aug 2025
Huber+Suhner (HUBN)	Polatis OCS	No evidence	No evidence	Major orders received July 2025

Source: Company data, Morgan Stanley Research. GA = General Availability

Regardless of which technology roadmap eventually succeeds, we believe OCS adoption could boost transceiver demand. We cite three reasons:

1) New deployment use cases creating additional demand

OCS is being deployed in multiple new applications within data centers, each requiring transceivers. According to Lumentum, there are three dominant use cases for OCS: spine switch replacement in greenfield deployments, TPU interconnect, and scale-up applications inside the rack itself. These new deployment scenarios create incremental demand for optical transceivers beyond traditional switching architectures.

2) Complementary rather than substitutive relationship

OCS and optical transceivers form a complementary ecosystem rather than competing technologies. While OCS provides the switching foundation, optical transceivers and modules are the building blocks that connect servers, switches, and clusters. This means OCS deployment doesn't replace transceivers but rather creates additional points where transceivers are needed.

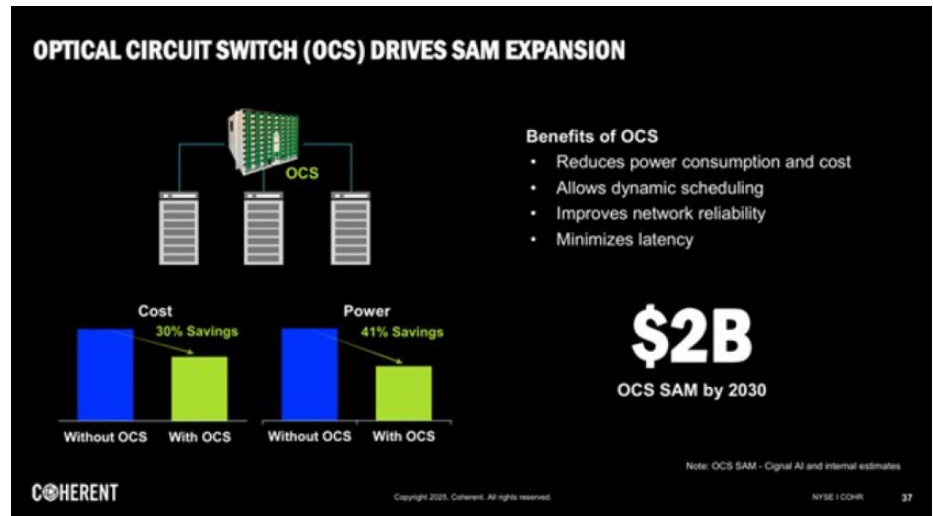
3) Higher performance requirements

OCS deployment imposes stricter requirements on transceiver performance, driving the industry toward higher data rates, lower insertion loss, and other enhanced specifications. The growing demand for 800G and 1.6T modules in AI and high-performance computing (HPC) clusters presents significant opportunities for transceiver manufacturers.

OCS could also introduce higher technology requirements for transceivers, including:

- Extended span budgets (up to approximately 6 km for OCS-interworked FR optics)
- Bi-directional operation using circulators
- A shift towards FR optics (single fiber) over DR optics (multiple fibers), favoring indium phosphide (InP)-based optics
- Higher output power and multi-wavelength options for seamless integration with OCS systems
- Higher power transceivers to handle the high insertion loss of OCS systems

The trend is accelerating as hyperscalers are following the OCS adoption trend, leading to a rapidly expanding market for OCS components and, consequently, the transceivers that support them.

Exhibit 14: Optical Vendors Banking on Seeing More OCS Expansion

Source: Coherent.

NPO – Traditional transceiver companies' best defense against CPO

NPO (near-packaged optics) transceivers are advanced optical modules in which the optical engine is moved from a traditional pluggable module to a position very close to the switch chip (ASIC) on the same board. This reduces power consumption and improves density by shortening the electrical path, often using silicon photonics for high-speed, efficient data center communication. NPO bridges the gap between pluggable modules and full co-packaged optics (CPO).

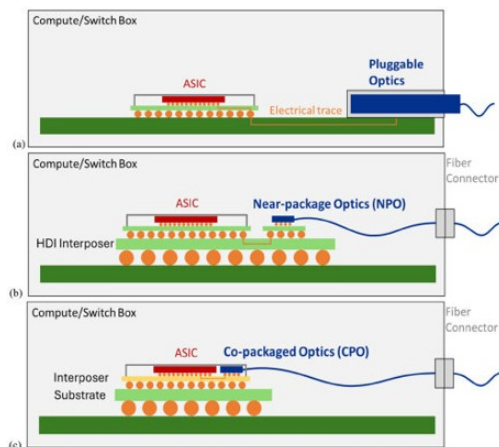
Key differences between NPO, CPO, and pluggables

- **Pluggable:** The optical engine is in a separate module, far from the ASIC, requiring more power.
- **NPO (near-packaged optics):** The optical engine is near the ASIC on the same board (e.g., <50mm away).
- **CPO (co-packaged optics):** The optical engine is integrated directly onto the same package as the ASIC, offering the highest level of integration.

As illustrated in [Exhibit 15](#), NPO sits on the same board as the ASIC; as a result, NPO performance is similar to CPO, with several advantages:

- **Short electrical path:** This proximity creates an extremely short electrical link (a few centimeters), significantly reducing signal loss and power consumption.
- **Power efficiency:** By shortening the path, the need for power-intensive digital signal processors (DSP) is reduced or eliminated, saving significant power (e.g., 800G transceivers using NPO can use much less power than traditional ones).
- **Performance:** Improved signal integrity allows much higher bandwidth and lower latency than traditional transceivers.

If traditional transceiver companies can achieve a breakthrough in NPO technology, they could offset potential disruption from CPO because NPO performance is similar to CPO – while manufacturing complexity is significantly reduced.

Exhibit 15: NPO vs. CPO and Pluggables

Source: Optics Express.

Significant revenue and earnings growth potential in 2026-2028

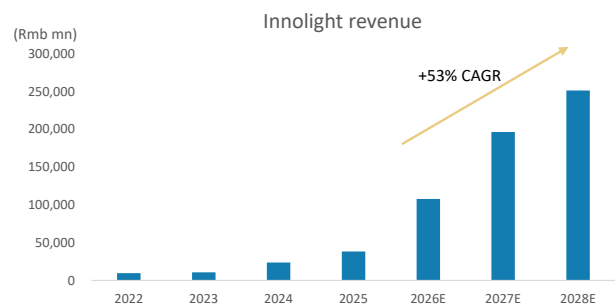
We believe InnoLight is well positioned for rapid revenue and earnings growth in 2026-28, driven by broad adoption of scale-out, scale-up, and scale-across technologies in data center architecture. The company benefits from strong demand for 800G and 1.6T transceivers. These new growth drivers have already supported the company's growth in 2025 and 1Q26. Looking ahead, both products are likely to support rapid growth in 2026, while 1.6T is more likely to become the key growth driver in 2027, supported by AI infrastructure upgrades.

We believe InnoLight's rapid growth in AI transceiver business is driven by 1) strong industry demand growth, and 2) the company's competitive advantages, which allow it to outperform peers in a) time-to-market, b) time-to-volume, and c) time-to-cost. Combining those two factors, we expect InnoLight to deliver superior top-line growth relative to the broader AI transceiver industry.

In addition, we expect NPO, OCS, and other potential new optical communication products to become new growth drivers from 2027 onward. The company's leading R&D capabilities and strong partnerships with top-tier CSP customers provide a strong foundation for capturing these new growth opportunities.

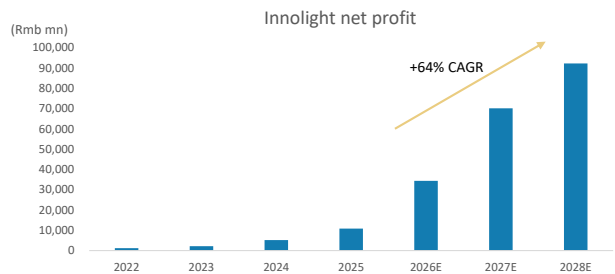
Combining the factors above, we forecast InnoLight to achieve a 53% revenue CAGR in 2026-28, while earnings could expand at 64% over the same period.

Exhibit 16: InnoLight's revenue growth, 2022-2028e



Source: Company data, Morgan Stanley Research (E) estimates.

Exhibit 17: InnoLight's earnings growth, 2022-2028e



Source: Company data, Morgan Stanley Research (E) estimates.

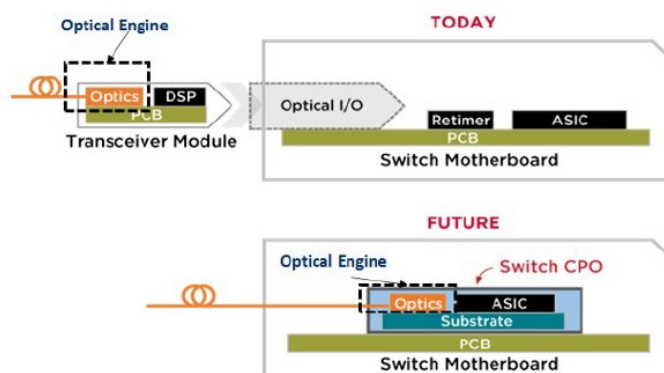
Investment Concerns

CPO development adds uncertainty to the technology roadmap and potential disruption risk

Co-packaged optics (CPO) is an advanced packaging technology that leverages silicon photonics. Its core idea is to integrate optical engines (such as optical transceiver modules) directly into the package of a switch chip or other processing chips. This can increase data transmission speeds, shorten transmission distance, reduce power consumption, and minimize latency.

CPO brings together expertise in fiber optics, digital signal processing (DSP), application-specific integrated circuit (ASIC) switches, and packaging and test. Together, these capabilities can create disruptive system value for data center and cloud infrastructure.

Exhibit 18: Migration from transceivers to CPO architecture



Source: Broadcom.

Why CPO could become a strategic challenge to traditional transceiver companies

CPO represents a fundamental architectural shift that poses a strategic challenge to traditional transceiver manufacturers. It integrates optical components directly into the switch or computing chip package, rather than using existing pluggable transceiver modules. If CPO technology development is successful, we expect three key risks for traditional transceiver companies:

1) Product displacement risk

CPO directly threatens the traditional pluggable transceiver business model. Unlike traditional pluggable transceivers, which connect to switches via electrical interfaces, CPO integrates optical transceivers directly into the same physical package as the switch ASIC or computing processor. This shift could lead to a significant decline in demand for traditional product lines.

2) Business model disruption

CPO requires new partnerships and a re-evaluation of business models. The technology involves highly complex design and manufacturing processes, requiring specialized equipment and expertise in silicon photonics and advanced packaging. This creates barriers for traditional transceiver manufacturers, which may lack these capabilities.

3) Competitive landscape shift

CPO competes not only against pluggable transceivers but also against copper in the data interconnect market. Amid these industry dynamics, copper technology is also evolving. Passive copper still dominates scale-up networks, and active copper is gaining share in scale-out networks. This creates additional competitive pressure on traditional optical transceivers.

However, CPO still faces several challenges

First, CPO faces several major technical challenges that create uncertainty for transceiver manufacturers, including:

- **Manufacturing yields:** Integrating optical components with silicon is technically demanding and currently results in low manufacturing yields.
- **Thermal management:** Integrating heat-sensitive optical components directly with high-power ASICs creates significant thermal challenges, often requiring advanced cooling solutions, like liquid cooling.
- **Cost premium:** CPO solutions today carry price points roughly 8-10 times higher than traditional pluggables.
- **Limited ecosystem:** There is scaled manufacturing capacity for tens of millions of transceivers, but the CPO ecosystem remains more limited and does not yet appear able to scale to comparable production volumes.

Second, CPO changes the failure domain from one port to the whole switch. With pluggable optics, if a port fails, the customer can replace that port and choose from multiple vendors. With CPO, the failure domain expands from 800 Gbps to 51 Tbps of bandwidth. This operational challenge makes customers cautious about the CPO paradigm because the "blast radius," or unit of failure, increases significantly. In addition, unlike hot-swappable pluggable transceivers, CPO modules are tightly integrated, making field replacement and failure management more complex.

Large-scale CPO adoption is likely in 2027 or later

Considering the technical challenges mentioned above, we believe CPO is unlikely to take share from traditional transceivers in the near term. It is likely to introduce competition only once large-scale application becomes more mature, which we believe is more likely in 2027 or later.

We also expect the impact to vary by network application scenario:

Scale-out networks: We believe CPO is likely to remain a contained offering rather than a broad-based network solution.

Scale-up networks: We believe CPO is more relevant for scale-up applications, where electrical connections have reached density and bandwidth limits and pluggables may not work. CPO will eventually shift the market away from DSP-based transceivers.

Combining the current state of CPO development with the expected timing of broad scale-up adoption, we believe meaningful CPO adoption is likely to begin in 2027-2028, with the initial crossover expected at 3.2T and beyond.

Morgan Stanley's views on CPO volume and timetable

The Morgan Stanley Research semiconductor team expects to see more CPO product launches and a more concrete development timeline [starting in 2H26](#), including both scale-up and scale-out applications.

At GTC 2025, NVIDIA launched its Quantum X800 and Spectrum-X CPO switches.

At CES 2026, NVIDIA elaborated more on Spectrum's product plan, which is considered the recommended Ethernet switch product alongside the new GPU Vera Rubin platform.

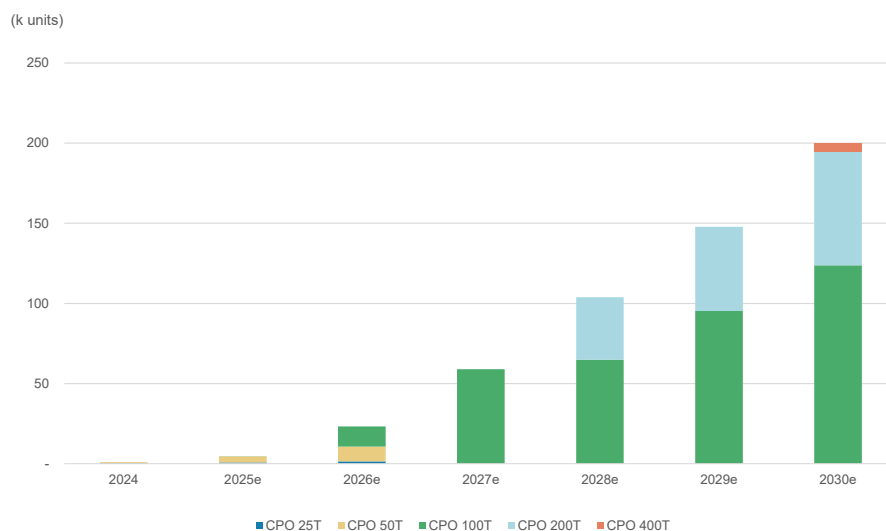
At GTC 2026, NVIDIA officially launched its Spectrum-X Photonics CPO switches and announced that CPO technology has moved into full, mass-production, marking a significant shift from pluggable optics to CPO for AI data centers. Key CPO-related announcements and developments at GTC 2026 included:

- Mass production of CPO switches: Jensen Huang confirmed that CPO hardware is in "full production" developed in partnership with TSMC.
- Spectrum-X photonics switches: NVIDIA launched Spectrum-X Photonics switches designed for agentic AI that integrate optical engines directly on to the ASIC. These switches, slated for release in the second half of 2026, enable up to 409.6 Tb/s bandwidth.
- Performance metrics: The CPO switches deliver 5x better power efficiency and 10x higher network reliability compared to traditional pluggable transceivers.
- CPO roadmap debut: CPO technology is making its debut in high-volume, scale-up networking with the new Rubin Ultra NVL576 and Feynman NVL1152 multi-rack systems, specifically for inter-rack and scale-out connections.
- CPO technology is designed to directly address the bandwidth and power consumption challenges in large-scale AI factories.

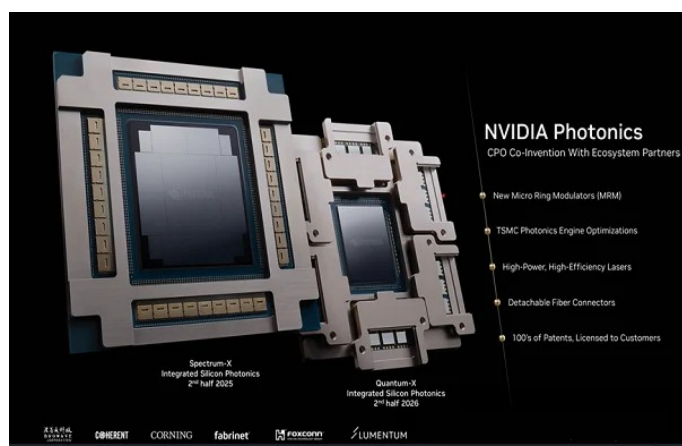
For 2026, we expect total CPO switch shipments to be 23k units, with the majority being 100T switches and NVIDIA's Spectrum switch taking the largest share.

Looking into 2027, we expect total CPO switch shipments to reach 59k units, growing further to 200k units in total in 2030e, which implies a 144% CAGR, 2024-2030e.

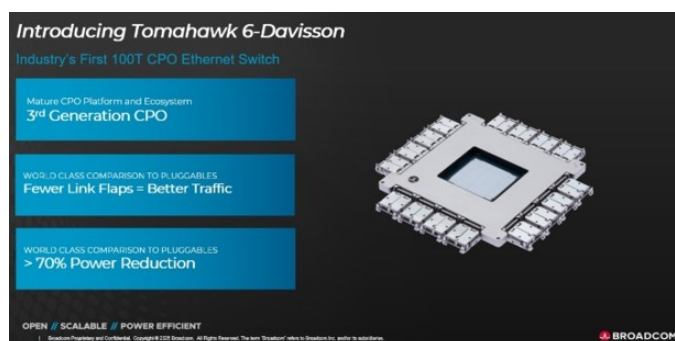
Our semiconductor team's supply chain checks suggest that TSMC plans to ramp-up PIC capacity from 10kwpm in 2Q26 to 15kwpm in 4Q26. To our knowledge TSMC has not confirmed this to date. They are seeing that its Silicon Photonics platform (COUPE) customers are requesting a very aggressive expansion schedule starting from 1Q27, which is partially to prepare for the scale-up or Optical I/O introduction. This points to the team's bull case scenario (or our bear case scenario for InnoLight) – in which CPO switch shipments are likely to reach 80k in 2026, 180k in 2027, and 250k in 2028.

Exhibit 19: We expect total CPO switch shipment CAGR to be 144%, 2024-2030e

Source: Yole, Morgan Stanley Research (E) estimates.

Exhibit 20: Quantum and Spectrum CPO switch

Source: NVIDIA.

Exhibit 21: Broadcom's latest CPO switch reaching 102.4T "Davisson"

Source: Broadcom.

Morgan Stanley Research bull-base-bear case scenario analysis to help gauge the risk from CPO disruption

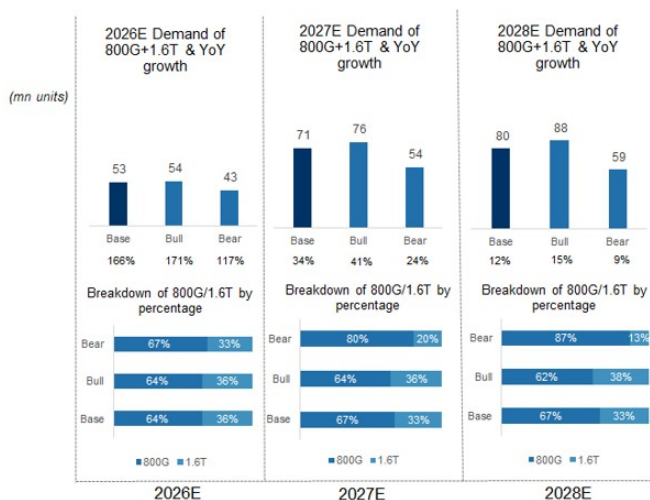
CPO is a significant potential disruptor for the traditional transceiver industry, but the timetable for large-scale application remains uncertain. We have therefore decided to apply bull-base-bear case scenario analysis to assess the fundamental impact.

Bull case: This scenario assumes that CPO disruption arrives after 2028, which could give sufficient time for traditional transceiver companies to develop 3.2T products with good quality and large-scale production. As a result, the traditional transceiver companies would be able to defend their market share with better customer experience and economics. We forecast traditional transceiver companies would take >70% high-end market share while CPO takes only <30% high-end market share by 2035.

Base case: This scenario assumes CPO achieves manufacturing breakthrough by 2027 with large-scale adoption beginning in 2H27 or 2028. As a result, CPO becomes the mainstream technology for next-generation high-end network connections. Under this scenario, even if traditional transceiver companies develop high-quality 3.2T products, they are less likely to maintain dominance in the high-end market. We forecast CPO would take >50% high-end market share while traditional transceiver companies hold <50% high-end market share by 2035.

Bear case: This scenario assumes CPO achieves a manufacturing breakthrough in 2H26 and 2027, with large-scale adoption beginning in 1H27. As a result, CPO disruption arrives earlier than expected. This development could make CPO an important driver of accelerated scale-up penetration. Under this scenario, we forecast CPO would take >70% high-end market share, while traditional transceiver companies hold <30% high-end market share by 2035.

Exhibit 22: Our base, bull, and bear case scenarios for 800G/1.6T global demand forecasts in 2026-2028



Source: Company data, Morgan Stanley Research (E) estimates.

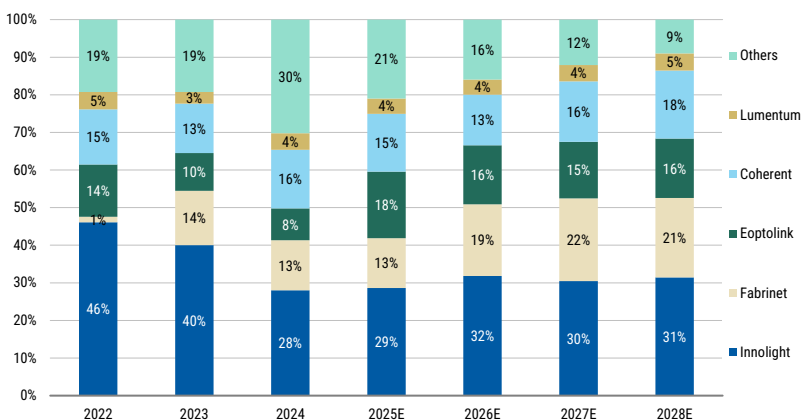
Rising competition as industry peers expand aggressively

While the industry is well positioned for rapid demand growth, our analysis suggests that all transceiver companies will expand capacity aggressively to capture this demand.

In addition, most CSPs are likely to diversify transceiver supply once order sizes reach a large scale. For example, during the initial launch period for 800G products, only companies such as InnoLight secured orders, allowing InnoLight to secure very high market share. However, as 800G becomes a mainstream product, at least five or six large transceiver companies are likely to produce 800G transceivers. As a result, InnoLight's market share is likely to decline relative to the initial stage.

We believe InnoLight will enjoy first-mover advantage for new products in the first 1-2 years, but it cannot prevent new players from joining supplier lists as products mature. As illustrated in [Exhibit 23](#), we expect industry peers such as Eoptolink and Coherent to gain some market share, while InnoLight is more likely to sustain its share via continuous new product innovation but cede some share to peers in mature product categories.

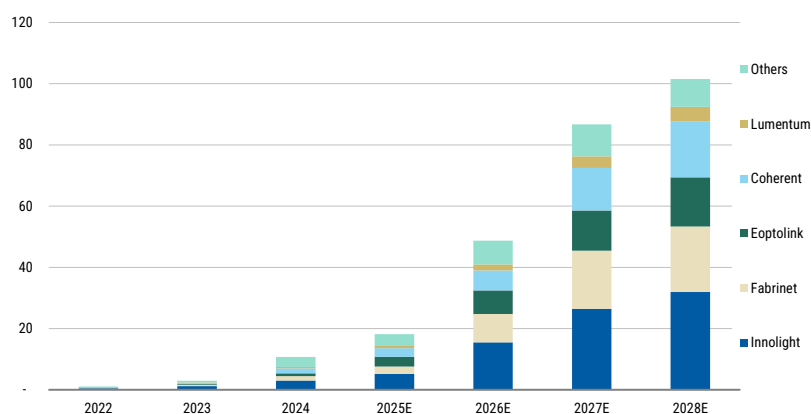
Exhibit 23: Global AI transceiver market share changes: Eoptolink and Coherent are likely to gain share



Source: Company data, Omdia, Morgan Stanley Research (E) estimates.

However, because total industry demand is likely to grow rapidly, we don't expect competition to become a zero-sum game. As illustrated in [Exhibit 24](#), InnoLight's stable market share won't prevent absolute volume/revenue growth. If new products have higher ASPs or better margins, the earnings impact could be more positive than the top-line impact.

Exhibit 24: Global AI transceiver TAM, by company (absolute amount)



Units in US\$ bn, Source: Company data, Omdia, Morgan Stanley Research (E) estimates.

Potential geopolitical tensions is a macro risk factor

Given InnoLight's large revenue exposure to overseas customers and its reliance on global supply chain partners, potential geopolitical tension is a macro risk factor for the company.

Since early 2025, particularly from April onward, the optical communications industry has faced global supply chain restructuring and geopolitical risks due to a series of tariff policies including additional tariffs and reciprocal tariffs. Although tariff exemption policies have subsequently been introduced, tariff policy volatility could adversely affect the company's operations. If US-China trade disputes escalate and trade sanctions deepen, this could disrupt transceiver demand, increase the difficulty of procuring key raw materials, and affect the company's profitability.

Because the whole industry is experiencing a shortage of some key components, such as EMLs, any geopolitical disputes could amplify supply chain risk, as InnoLight sources a significant portion of its key raw materials from overseas vendors.

To mitigate these risks, InnoLight has taken measures including 1) continuing to enlarge its overseas manufacturing footprint and further optimize its supply chain and production capacity layout to enhance its ability to respond to various risk events, and 2) closely monitoring tariff policy changes and fully researching and using tariff exemption policies to seek favorable development opportunities amid ongoing policy adjustments.

On [August 4, 2026](#), [Reuters reported](#) that Washington is moving to block new Chinese optical transceivers from U.S. data centers, with the FCC-led measure citing fears of espionage, malware, or sabotage amid the AI race. As of this report's date, no formal rule had been published. The White House and FCC did not formally comment, while the Chinese embassy in the U.S. voiced objections to the reported proposal. Any action appears more likely to restrict future product approvals than installed equipment, with the scope across Chinese components and non-Chinese modules still unclear.

While no formal rule has yet been published, we believe this latest development illustrates potential risk from geopolitical tensions.

We believe the scope of "Chinese" origin (mainland China vs. offshore subsidiaries) is critical for the real impact. If the regulation only takes effect for mainland China capacity, it would have only very limited fundamental impact given InnoLight has already established overseas production. However, if the offshore subsidiaries are also included, it could introduce significant impact as InnoLight has large revenue exposure to US CSP customers.

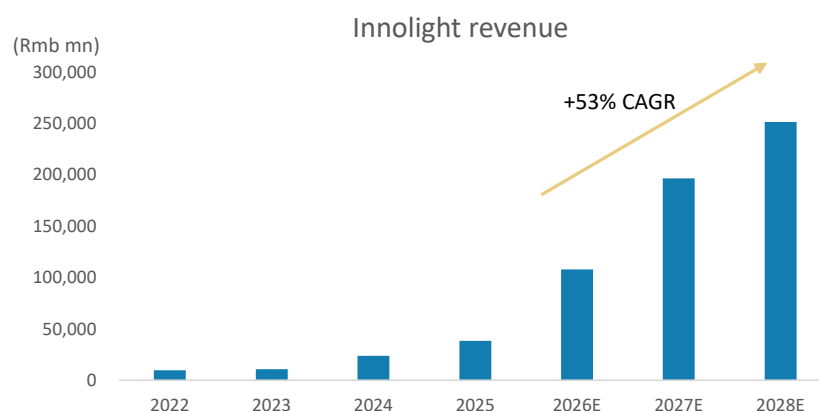
However, we believe such regulation is unlikely to take effect given the non-China suppliers currently lack the manufacturing capacity to replace the volume provided by Chinese companies. Any disruption of the AI transceiver supply could lead to capacity deficits and significant delay of AI infrastructure build-out in the US. We believe the risk could increase if there is more aggressive capacity expansion by other non-China suppliers. Without such additional capacity, the risk of stricter regulation is low, in our view.

Earnings Outlook and Financial Forecasts

Revenue growth outlook

InnoLight's revenue rose at a 58% CAGR, from Rmb9.6bn in 2022 to Rmb38.2bn in 2025, driven by rapid shipment growth and ASP increases. We expect rapid end-demand expansion for high-end optics and company-specific execution advantages to drive strong revenue growth in 2026-2028. We forecast InnoLight's revenue to reach a respective Rmb107.9bn / Rmb196.3bn / Rmb251.3bn in 2026 / 2027 / 2028, representing a 53% CAGR.

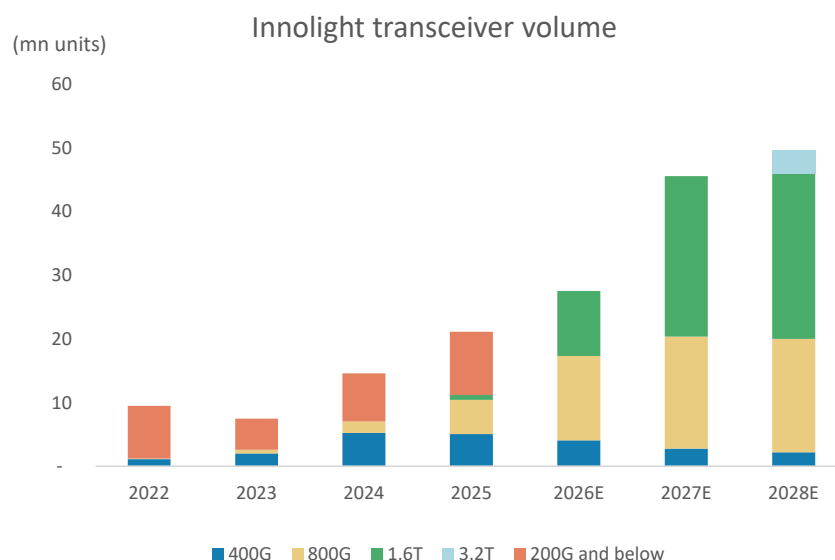
Exhibit 25: InnoLight revenue forecasts, 2022-2028



Source: Company data, Morgan Stanley Research (E) estimates.

Transceiver volume assumptions

Fueled by 800G demand, InnoLight's shipment volume grew at a 31% CAGR, from 9.46mn units in 2022 to 21.09mn units in 2025. We expect 1.6T to be the main growth driver in 2026-2028, with InnoLight leading the market with a market share of 30-35%. We also expect 3.2T to become a new growth engine in 2028. As a result, we forecast InnoLight's total shipment volume to reach 27.5mn / 45.5mn / 49.5mn units in 2026 / 2027 / 2028, respectively, representing a 34% CAGR.

Exhibit 26: InnoLight transceiver volume assumptions

Source: Company data, Morgan Stanley Research (E) estimates.

Transceiver ASP assumptions

Driven by product upgrades and mix improvement, InnoLight's ASP grew at a 22% CAGR, from US\$136 in 2022 to US\$245 in 2025. With a rising contribution from 800G and 1.6T products, we forecast InnoLight's ASP to rise at a further 7% CAGR, reaching US\$566 / US\$582 / US\$645 in 2026 / 2027 / 2028, respectively.

Exhibit 27: InnoLight ASP assumptions

ASP (US\$)	2022	2023	2024	2025	2026E	2027E	2028E
400G	400	370	356	328	304	285	271
800G	800	750	650	480	447	402	374
1.6T			1,673	1,171	820	738	701
3.2T							1,800
Average ASP	136	189	220	245	566	582	645
YoY change							
400G		-8%	-4%	-8%	-8%	-6%	-5%
800G		-6%	-13%	-26%	-7%	-10%	-7%
1.6T				-30%	-30%	-10%	-5%
Average ASP		39%	16%	12%	131%	3%	11%

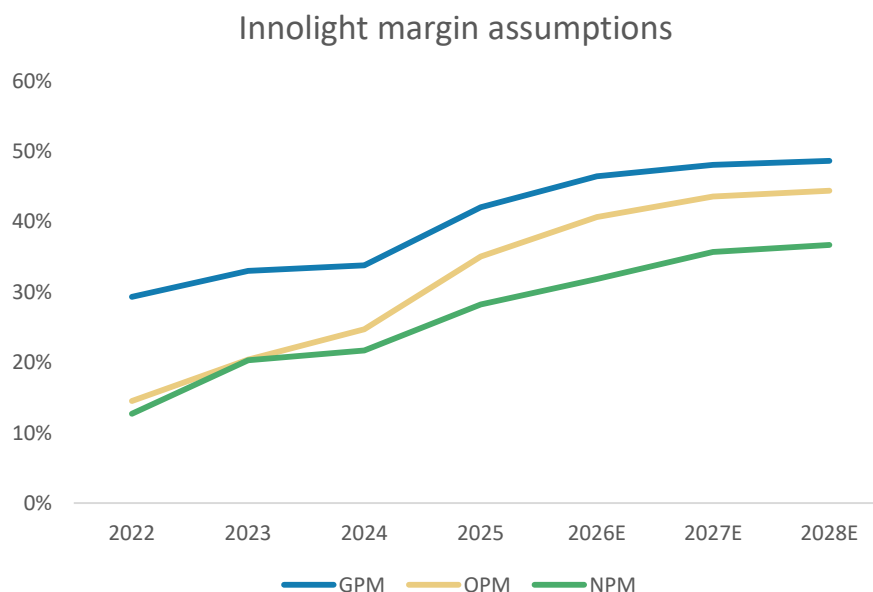
Source: Company data, Morgan Stanley Research (E) estimates.

Margin outlook

InnoLight's GPM quickly improved from 29.3% in 2022 to 42% in 2025, supported by an improved product mix with higher-margin products. We expect margin expansion to continue through 2026-2028, as InnoLight optimizes its mix and brings new, higher-ASP products to market. As a result, we expect GPM to expand further, to a respective 46.4% / 48.1% / 48.6% in 2026 / 2027 / 2028.

OPM and NPM have followed similar trends to GPM. We expect InnoLight's OPM to improve to a respective 40.7% / 43.6% / 44.4% in 2026 / 2027 / 2028. As a result, we expect NPM to improve to a respective 31.8% / 35.7% / 36.7% in 2026 / 2027 / 2028.

Exhibit 28: InnoLight GPM, OPM, and NPM, 2022-2028

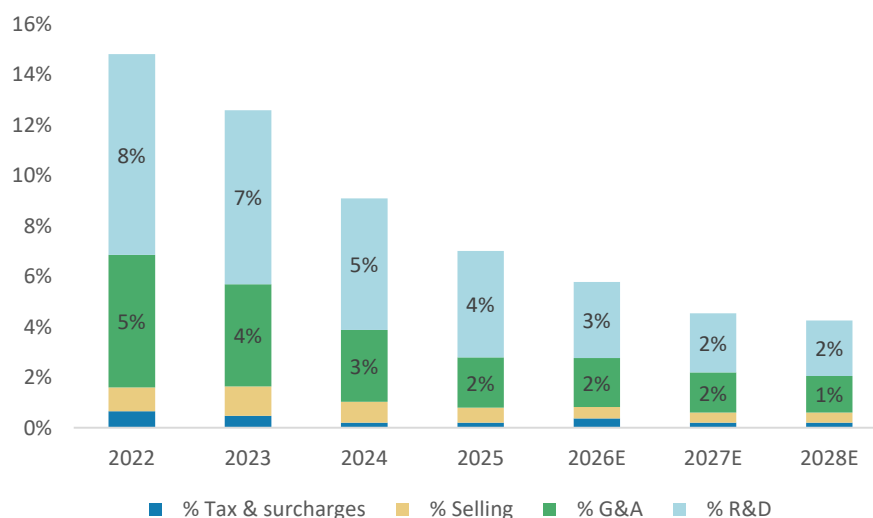


Source: Company data, Morgan Stanley Research (E) estimates.

OPEX assumptions

InnoLight's opex ratio declined from 14.8% in 2022 to 7.0% in 2025, driven by scale efficiencies. We project further reductions in the opex ratio during 2026-28, underpinned by continued scale benefits and lower administration expenses. We forecast the opex ratio to decline to a respective 5.8% / 4.5% / 4.2% in 2026 / 2027 / 2028.

Exhibit 29: InnoLight OPEX ratio, 2022-2028

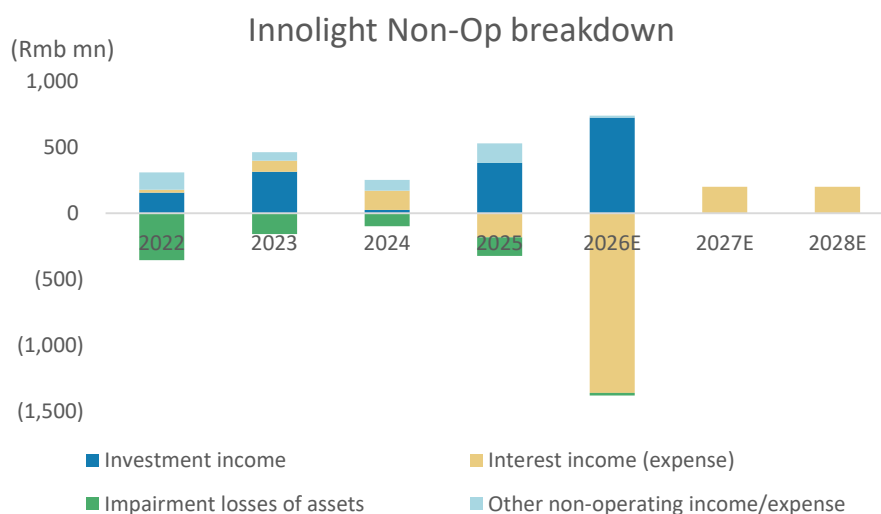


Source: Company data, Morgan Stanley Research (E) estimates.

InnoLight's non-operating income mainly includes investment income, interest income/expense and impairment losses. Given its high exposure to overseas markets, the company is heavily affected by RMB exchange rate fluctuations. The FX gain/loss is also recorded in this category. If there is an FX loss, it will be recorded as a financial expense. In 1Q26, the company recorded a financial expense of Rmb250mn due to FX loss. As our [China equity strategy team expects RMB appreciation through end-2026](#), we conservatively forecast the financial expense to remain high in the next few quarters. We thus forecast the financial expense to rise to Rmb1.36 bn in 2026, resulting in Rmb643mn of non-operating expenses.

However, we believe this impact is more likely to be one-off, as the RMB would not appreciate continuously. For 2027-2028e, we assume the FX rate to remain stable; therefore, the company is less likely to incur large financial cost. Meanwhile, we believe the company will continue to record rising interest income with higher cash balance. As a result, normalized non-operating income could return to positive territory, and we estimate it at Rmb200mn scale per annum.

Exhibit 30: InnoLight non-OP breakdown, 2022-2028

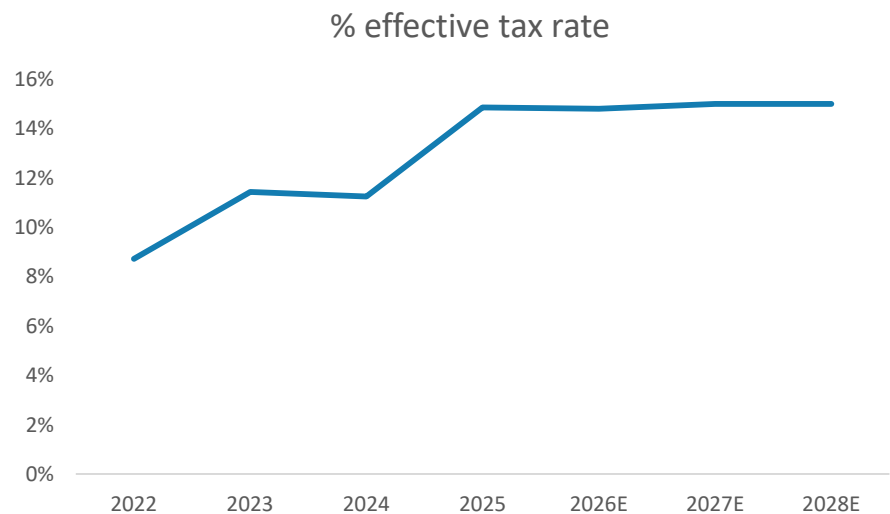


Source: Company data, Morgan Stanley Research (E) estimates.

Tax rate

InnoLight's effective tax rate increased from 8.7% in 2022 to 14.9% in 2025. We expect InnoLight to maintain a relatively stable tax rate of 14.8% / 15% / 15% in 2026 / 2027 / 2028, respectively.

Exhibit 31: InnoLight effective tax rate, 2022-2028

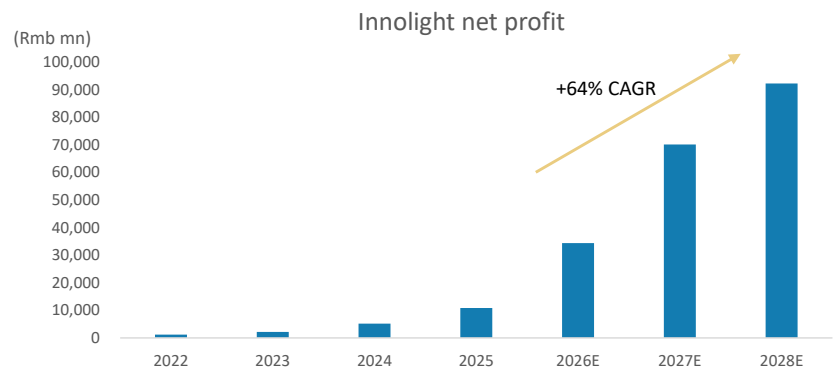


Source: Company data, Morgan Stanley Research (E) estimates.

Net profit

As a result of sustained market share gains, product mix improvement, and efficiency gains, we project InnoLight's net profit to reach Rmb34.3bn / Rmb70.1bn / Rmb92.2bn in 2026 / 2027 / 2028, respectively, representing a 64% CAGR.

Exhibit 32: InnoLight earnings forecasts, 2022-2028



Source: Company data, Morgan Stanley Research (E) estimates.

Key risks to our financial forecasts

Upside risks: InnoLight's earnings and valuation could surprise to the upside if AI-driven data-center capex proves stronger or more durable than expected, accelerating adoption of 800G and 1.6T optical modules and extending the industry upcycle. Faster-than-anticipated penetration of silicon products, combined with successful customer qualification and share gains at hyperscale clients, could support higher volumes, a better product mix and margin uplift. Additional upside could come from slower-than-expected CPO disruption after 2028, success with NPO, improved yields, and cost reductions.

Downside risks: Key downside risks include a slowdown or volatility in hyperscale capex, delayed deployment of next-generation optical modules, or aggressive pricing competition that compresses margins. High customer concentration increases exposure to procurement shifts or dual-sourcing decisions, while yield challenges or cost inflation could weigh on profitability. In addition, a CPO manufacturing breakthrough in 2027 could bring disruption earlier than expected. Geopolitical factors, export controls, and broader macro weakness also pose risks to demand visibility and international sales.

Exhibit 33: 800G to fuel material growth beginning in 2024-2025, with 1.6T ramping a few years from now

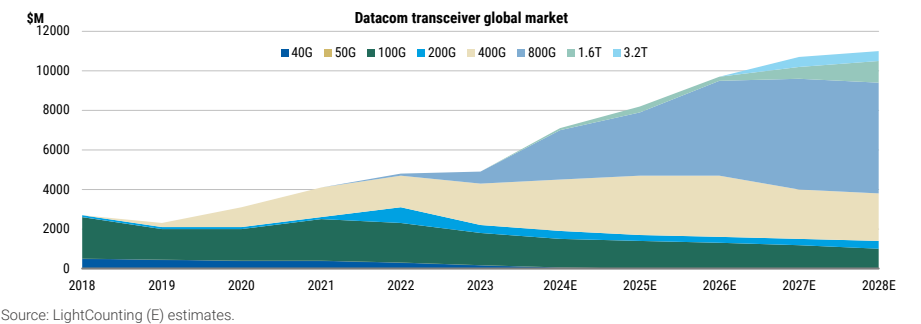
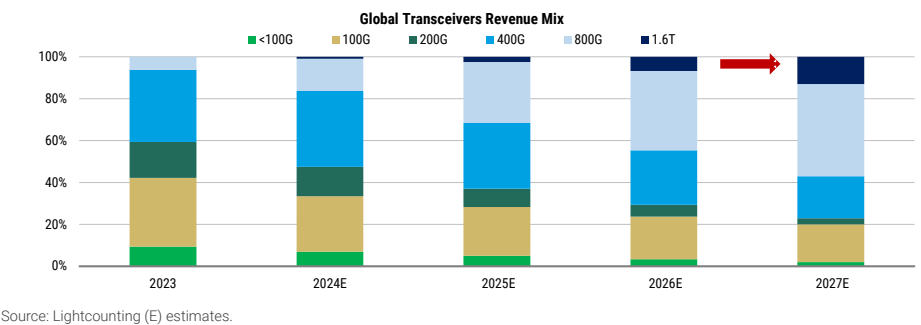


Exhibit 34: Global transceiver revenue mix



Valuation

Considering the high uncertainty about whether and when CPO breakthrough can be achieved, we have decided to apply a probability-weighted bull-base-bear valuation methodology (30% bull, 50% base, 20% bear, with the skew reflecting the expected timeline for large-scale CPO application). In each scenario, we use a residual income (RI) model to derive the intrinsic value. The growth, margin, and return estimates are different in each scenario, but some key assumptions are the same across all three scenarios, including 1) 10% cost of equity, 2) 3.5% terminal growth rate, and 3) 1.15 RMB/HKD FX rate.

Exhibit 35: InnoLight: Base case residual income model

InnoLight (3308.HK)
InnoLight (300308.SZ)

Residual Income Valuation

	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Total equity	107,555	172,663	254,759	271,230	290,172	311,387	335,148	361,285	389,513	419,576	451,563
Total revenue	107,904	196,339	251,279	288,971	332,317	372,195	416,858	458,544	495,227	527,417	561,177
Net Profit	34,346	70,056	92,189	86,691	99,695	111,658	125,057	137,563	148,568	158,225	168,353
Return on Equity	115.4%	65.1%	53.4%	34.0%	36.8%	38.5%	40.2%	41.0%	41.1%	40.6%	40.1%
Residual Income	31,369	59,300	74,922	61,215	72,572	82,641	93,919	104,048	112,440	119,274	126,395
Equity Value (Rmb bn)	1,404										
Equity Value (HKD bn)	1,614										
No. of Shares	1,170										
Projected Price (EoY Rmb)	1,201										
Projected Price (EoY HKD)	1,380										
P/E (x)	40.9	20.0									
P/B (x)	12.4										

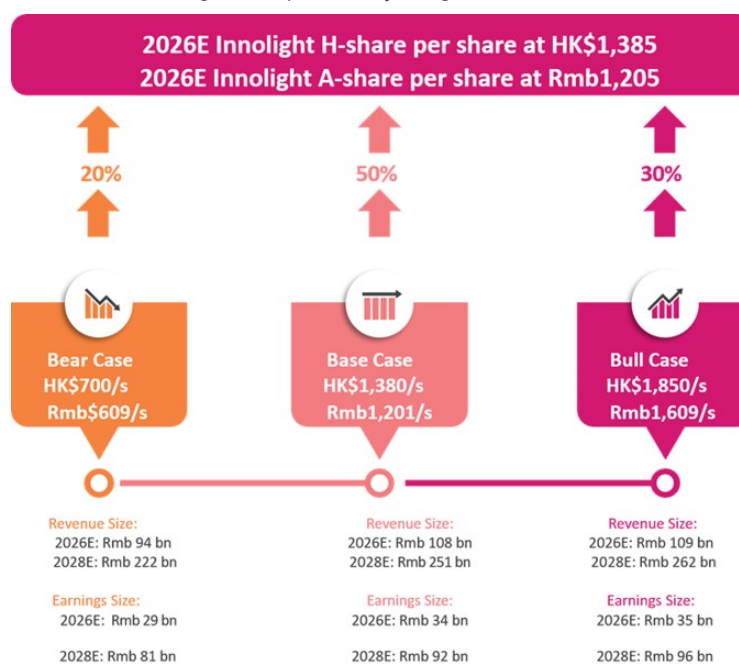
Source: Company data, Morgan Stanley Research (E) estimates.

In our base case scenario, we assume the CPO breakthrough will arrive in 2027 or later, while the company's revenue increases from Rmb108bn in 2026 to Rmb251bn in 2028, with earnings increasing from Rmb34bn to Rmb92bn over the same period. After that, the earnings CAGR could reach 8% in 2028-2036e.

In our bull case scenario, we assume the CPO breakthrough is further delayed while NPO technology becomes mature ahead of that. As a result, the company's revenue increases from Rmb109bn in 2026 to Rmb262bn in 2028, with earnings increasing from Rmb35bn to Rmb96bn over the same period. After that, the earnings CAGR could reach 12% in 2028-2036e.

In our bear case scenario, we assume CPO achieves earlier-than-expected breakthrough; as a result, the company's revenue grows more slowly, from Rmb94bn in 2026 to Rmb222bn in 2028, with earnings increasing from Rmb29bn to Rmb81bn over the same period. After that, the earnings CAGR could be 1% in 2028-2036e.

Our probability-weighted bull-base-bear valuation yields an intrinsic value of Rmb1,205/share for the A-share, and, based on a 1.15 RMB/HKD FX rate, an H-share price target of HK \$1,385/share.

Exhibit 36: InnoLight: Our probability-weighted bull-base-bear valuation

Source: Morgan Stanley Research estimates.

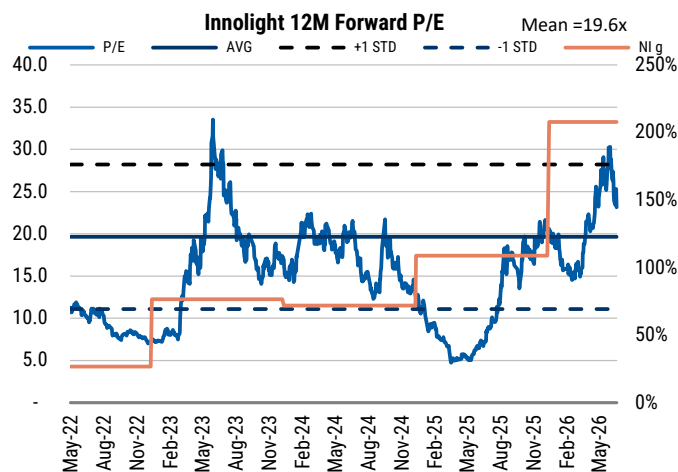
Exhibit 37: InnoLight: Our probability-weighted bull-base-bear valuation

InnoLight			
	Weight	RI Value (Rmb)	RI Value (HKD)
Base Case	50%	1,201	1,380
Bull Case	30%	1,609	1,850
Bear Case	20%	609	700
Weighted Avg. RI Value	100%	1,205	1,385

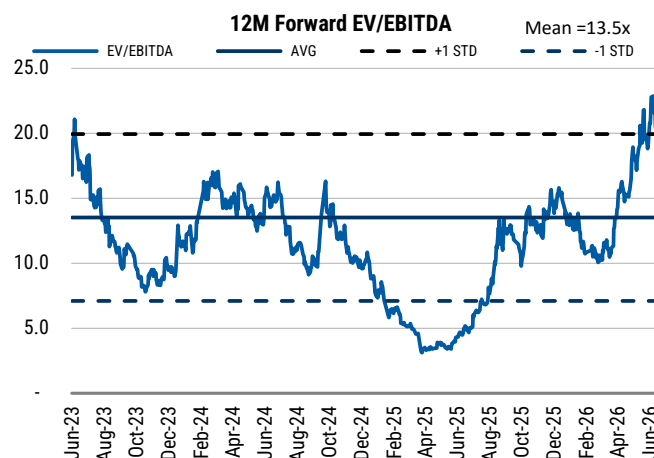
Source: Morgan Stanley Research estimates.

Forward P/E valuation range

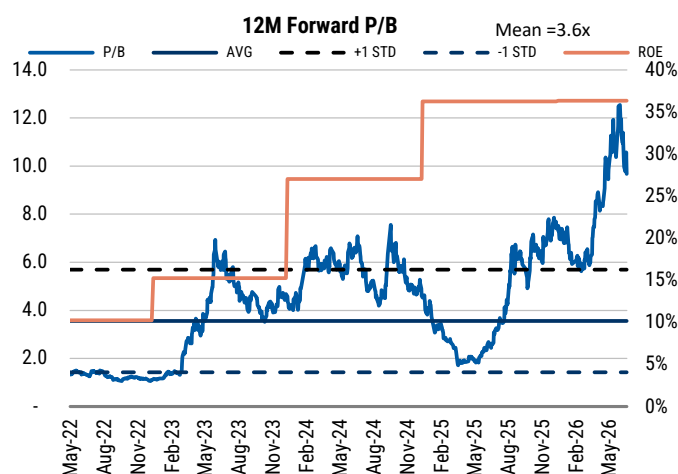
Reflecting the rising share price of InnoLight, valuations have increased. For example, since the beginning of 2025, InnoLight-A's 12-month forward P/E has risen from 13x to 23x. If we compare current valuations to historical ranges, InnoLight-A is trading at slightly above its +1 standard deviation level.

Exhibit 38: InnoLight: Forward P/E

Source: FactSet, company data, Morgan Stanley Research estimates.

Exhibit 40: InnoLight: Forward EV/EBITDA

Source: FactSet, company data, Morgan Stanley Research estimates.

Exhibit 39: InnoLight: Forward P/B

Source: FactSet, company data, Morgan Stanley Research estimates.

Comparison with other industry peers

InnoLight's H-share is trading at 14.4x 2027 P/E, on our estimates. Compared with global industry peers' current average P/E of 34.4x in 2027, InnoLight-H is trading at a 58% discount.

InnoLight's A-share is trading at 13.6x 2027 P/E, on our estimates. Compared with A-share industry peers' current average P/E of 53.3x in 2027, InnoLight-A is trading at a 75% discount.

We believe Eoptolink is the closest comparable company. Eoptolink is the second-largest transceiver supplier in China and, like InnoLight, primarily serves large global CSP customers. In addition, both companies have established manufacturing bases overseas. In new products such as 1.6T transceivers, InnoLight has a 6- to 12-month advantage, although Eoptolink could narrow the gap over the next year. Eoptolink currently trades at 16.0x 2027 P/E, on our estimates.

Our InnoLight price target implies 20.1x 2027e P/E, slightly below Eoptolink's 21.3x 2027e P/E implied by its price target. We believe this valuation discount reflects our more conservative stance on InnoLight. Industry leaders often trade at a premium to peers; therefore, InnoLight's discount to Eoptolink suggests that the current valuation already reflects potential risks, including component shortages and CPO disruption, which could weigh on the company's outlook.

Exhibit 41: InnoLight: Valuation Comp Sheet

Company	Ticker	Coverage	CURR	Last px (LoC)	Target Price (LoC)	Upside	MS Rating	Mkt Cap (USD Bn)	EV (USD Bn)	Total Return			52-wk		P/E(x)				P/S (x)				
Subject Company										1-mo	3-mo	12-mo	High	Low	25A	26E	27E	28E	25A	26E	27E	28E	
Zhongji InnoLight Co., Ltd. Class A	300308-CN	300308.SZ	Y	CNY	813.0	1,205	48%	OW	142	141	-10	-36	91	1,416.9	337.0	83.7	27.7	13.6	10.3	25.0	8.9	4.9	3.8
Zhongji InnoLight Co., Ltd. Class H	3308-HK	3308.HK	Y	HKD	995.0	1,385	39%	OW	142	141	NM	NM	NM	1,319.0	880.0	89.0	29.5	14.4	11.0	25.4	9.0	4.9	3.9
A-share peers																							
Eoptolink Technology Inc., Ltd. Class A	300502-CN	300502.SZ	Y	CNY	384.8	507.14	32%	OW	80	104	-2	-31	46	618.9	205.2	56.2	28.9	16.0	13.2	21.6	9.9	5.8	4.8
Suzhou TFC Optical Communication Co., Ltd. Class A	300394-CN	300394.SZ	Y	CNY	248.8	265.00	7%	EW	40	40	38	-30	70	375.0	100.8	134.2	78.8	56.7	43.6	52.6	26.4	18.6	14.1
Accelink Technologies Co., Ltd. Class A	002281-CN	002281.SZ	Y	CNY	173.5	166.00	-4%	UW	21	20	11	-23	165	279.0	54.0	147.9	75.4	51.3	38.3	12.0	9.3	6.9	5.2
T&S Communications Co., Ltd. Class A	300570-CN	300570.SZ	NC	CNY	188.6	NA	NA	NC	6	6	59	12	49	285.9	86.0	142.9	79.2	47.3	31.6	NA	16.3	9.3	5.9
Robotechnik Intelligent Technology Co., Ltd. Class A	300757-CN	300757.SZ	NC	CNY	585.2	NA	NA	NC	15	15	42	-15	146	714.0	189.2	NA	784.4	122.8	118.9	NA	54.1	14.0	6.5
Yangtze Optical Fibre & Cable Joint Stock Ltd. Co. Class A	601869-CN	601869.SS	Y	CNY	393.5	202.25	-49%	UW	33	32	50	-12	332	600.0	72.3	400.3	43.4	25.6	28.5	15.4	8.0	6.4	5.5
Linktel Technologies Co., Ltd. Class A	301205-CN	301205.SZ	NC	#REF!	285.0	NA	NA	NC	1	6	12	-15	148	378.0	86.2	360.8	NA	NA	NA	NA	NA	NA	NA
Average																207.0	181.7	53.3	45.7	25.4	20.6	10.2	7.0
Global peers																							
Lumentum Holdings, Inc.	LITE-US	LITE.O	Y	USD	847.4	1,000.00	18%	EW	76	72	9	-10	532	1,085.7	131.0	412.3	97.7	40.2	24.5	46.2	25.2	12.2	7.6
Coherent Corp.	COHR-US	COHR.N	Y	USD	264.4	375.00	42%	EW	52	61	-8	-37	199	440.0	87.4	74.8	47.1	28.6	19.9	8.9	7.3	4.9	3.6
Average																216.2	154.4	48.6	39.8	26.1	19.5	9.8	6.6
Industry Average																							

Company	P/B (x)				ROE (%)				EV/EBITDA (x)				Dividend Yield (%)											
Subject Company	25A	26E	27E	28E	25A	26E	27E	28E	25A	26E	27E	28E	25A	26E	27E	28E	25A	26E	27E	28E				
Zhongji InnoLight Co., Ltd. Class A	30.3	8.4	5.3	3.6	36%	32%	41%	36%	66.1	19.6	9.7	6.8	0.2%	0.5%	1.1%	1.4%								
Zhongji InnoLight Co., Ltd. Class H	32.3	9.0	5.6	3.8	36%	32%	41%	36%	70.5	21.2	10.5	7.6	0.2%	0.5%	1.0%	1.3%								
A-share peers																								
Eoptolink Technology Inc., Ltd. Class A	29.6	15.1	8.0	5.2	114%	103%	94%	60%	47.6	21.1	12.5	9.7	0.1%	0.2%	0.4%	0.7%								
Suzhou TFC Optical Communication Co., Ltd. Class A	49.2	32.3	22.1	15.8	51%	63%	57%	51%	99.2	59.9	43.7	33.3	0.2%	0.3%	0.5%	0.6%								
Accelink Technologies Co., Ltd. Class A	13.8	12.1	10.2	8.5	10%	18%	24%	27%	80.8	49.2	35.2	26.2	0.2%	0.3%	0.4%	0.6%								
T&S Communications Co., Ltd. Class A	25.1	21.3	16.4	12.2	18%	24%	36%	44%	112.4	70.1	36.5	21.5	0.3%	0.4%	0.5%	0.6%								
Robotechnik Intelligent Technology Co., Ltd. Class A	NA	57.1	49.2	34.8	-5%	14%	91%	153%	1,627.7	249.4	43.9	19.0	0.0%	0.1%	0.6%	2.0%								
Yangtze Optical Fibre & Cable Joint Stock Ltd. Co. Class A	NA	NA	NA	NA	5%	42%	51%	32%	58.2	15.9	10.1	10.6	0.1%	0.1%	0.7%	1.2%								
Linktel Technologies Co., Ltd. Class A	22.8	20.2	16.4	12.8	7%	13%	20%	23%	211.6	87.2	51.9	39.3	NA	NA	NA	NA								
Average	28.1	26.3	20.4	14.9	29%	40%	53%	56%	319.6	79.0	33.4	22.8	0.1%	0.2%	0.5%	0.9%								
Global peers																								
Lumentum Holdings, Inc.	52.0	23.8	18.3	12.5	-3%	52%	43%	53%	288.1	61.6	21.6	12.1	NA	NA	NA	NA								
Coherent Corp.	6.9	5.9	5.4	4.6	10%	15%	20%	27%	45.3	32.3	18.9	12.7	NA	NA	NA	NA								
Average	29.4	14.9	11.9	8.6	3%	34%	31%	40%	166.7	47.0	20.2	12.4	NA	NA	NA	NA								
Industry Average	28.5	23.5	18.3	13.3	23%	38%	48%	52%	285.6	71.9	30.5	20.5	0.1%	0.2%	0.5%	0.9%								

Source: FactSet, company data, Morgan Stanley Research. Morgan Stanley Research estimates for covered companies, FactSet consensus estimates for Non-Covered (NC) companies. Note: Latest share price is as the closing date of Sep 3, 2026.

Industry Overview

The AI transceiver industry is entering an exponential growth phase. We expect the industry TAM to increase from approximately US\$18 billion in 2025 to US\$102 billion by 2028, expanding by more than 4x within three years. In parallel, we project global AI transceiver demand to rise from approximately 20 million units in 2025 to 150 million units by 2028.

We also expect new growth opportunities, such as OCS and NPO, to become two additional technological breakthroughs in the coming years. If these technologies break through earlier than expected, we would expect a stronger positive impact on the transceiver industry, supporting the industry bull case scenario.

Exhibit 42: Industry demand forecasts

Base Case	2026e	2027e	2028e	CAGR 2026–2028e
800G	44	63	64	20%
1.6T	29	79	87	72%
Total	73	141	150	43%
Bull Case	2026e	2027e	2028e	CAGR 2026–2028e
800G	45	64	64	20%
1.6T	30	82	94	78%
Total	74	146	158	46%
Bear Case	2026e	2027e	2028e	CAGR 2026–2028e
800G	39	59	61	25%
1.6T	24	66	68	67%
Total	63	124	129	43%

Source: Morgan Stanley Research (E) estimates.

Google's latest fundraising implies AI investment is accelerating, not plateauing

On June 1, 2026, Alphabet Inc. announced what is considered its largest equity fundraising in history – an equity offering totaling US\$80 billion in expected aggregate amount, primarily to fund investments in AI compute infrastructure. Due to higher-than-expected investor demand, the offering was subsequently upsized and priced at US\$84.75 billion on June 2, 2026. The funds are intended to support the expansion of data centers, computing capacity, and systems for training and deploying large language models.

We believe such milestone fundraising has the following implications for AI infrastructure spending:

1) Unprecedented scale of hyperscaler capex: Alphabet's raise is part of a broader hyperscaler trend. Capex spend on AI infrastructure has grown over 133%, with hyperscalers already announcing plans to increase capex by over 70% in 2026. Alphabet's move signals that this wave of investment is accelerating, not plateauing.

2) Demand exceeds supply – a structural constraint: Alphabet explicitly stated that demand for its AI solutions and services from enterprises and consumers is "at levels that are exceeding the company's available supply." This compute-constrained environment is the primary driver of the capital raise and underscores the urgency of infrastructure buildout globally.

3) New financing models for AI infrastructure: Alphabet's multipronged approach – combining equity, debt, and operating cash flow – sets a template for how large-scale AI infrastructure can be financed. The equity raise is complemented by approximately US\$20 billion in additional debt issuance since Q1 2026.

4) Public-private partnerships and JVs: Alphabet is also expanding AI infrastructure through partnerships. In May 2026, Blackstone and Google formed a US-based joint venture with a US\$5 billion initial equity commitment from Blackstone to provide AI-focused data center capacity powered by Google's TPUs, with the first 500 megawatts of capacity expected online in 2027.

5) Global geographic expansion: Google has committed to investing US\$15 billion over five years to build an AI data center hub in Visakhapatnam, India – its largest investment in the country – combining gigawatt-scale data center operations, new energy sources, and expanded fiber-optic networks. Separately, Google announced a US\$4 billion investment in a new AI-focused data center in West Memphis, Arkansas.

6) Competitive moat and barriers to entry: The sheer scale of investment – with 2027 capex expected to exceed 2026's already record number – creates a structural advantage for hyperscalers. Smaller competitors and new entrants will face significant challenges replicating this infrastructure base, concentrating AI compute capacity among a handful of global players.

In summary, Alphabet's record-breaking equity raise is both a reflection of, and a catalyst for, the global AI infrastructure investment supercycle. It validates the thesis that AI compute is a strategic, capital-intensive asset class, and signals that the race to build foundational AI infrastructure will intensify significantly through 2027 and beyond.

AI Transceiver Growth Forecasts

We forecast global high-end (800G+1.6T) AI transceiver demand to rise from approximately 20 million in 2025 to 150 million by 2028. According to LightCounting, 800G mass production started in 2024 and 1.6T commercial production began in 2025, with 1.6T already entering scale production in 2H25 following the resolution of GPU delivery delays. The 3.2T generation is also in the R&D pipeline and could become another new growth driver in the years ahead.

Key Growth Drivers:

1) AI Data Center Expansion

The rapid increase in demand for AI data centers is the primary driver of transceiver growth. As AI workloads expand, connectivity is moving away from copper and electrical systems toward optical content, requiring advanced optical transceivers for rack-to-rack connections.

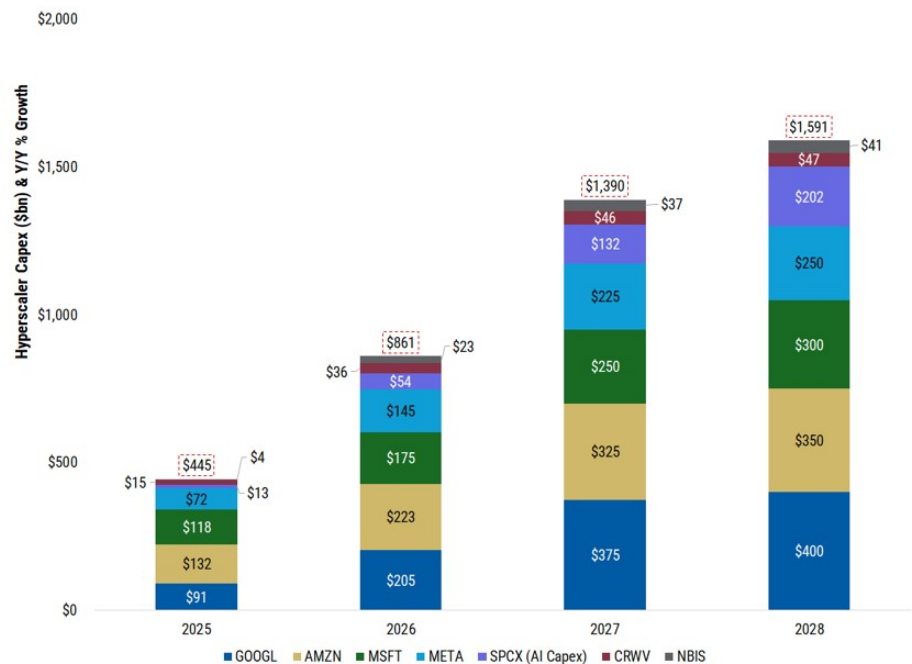
2) Surging Capital Expenditures from CSPs

Our US team now estimates total hyperscaler cash capex at approximately US\$861 billion in 2026, representing ~93% YoY growth. Key examples include:

- Amazon: ~US\$223 billion in capex in 2026 (+69% YoY).
- Google/Alphabet: Guided for 2026 capex of ~US\$205 billion (+124% YoY).
- Microsoft: Guided for capex of ~US\$175 billion for FY26 (+48% YoY).
- Meta Platforms: Guided for 2026 capex of ~US\$145 billion (+100% YoY).

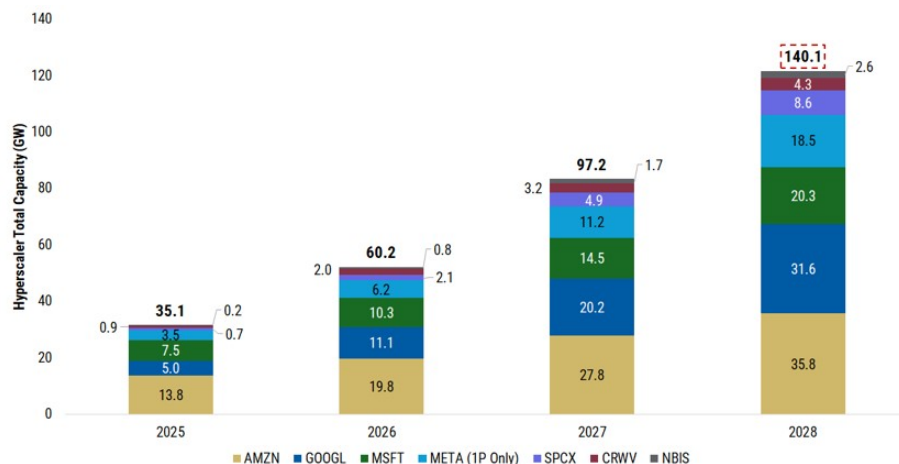
We believe that capex growth momentum will be sustained in 2027-2030, given bottlenecks in AI computing power, which require further AI infrastructure investment.

Exhibit 43: Our US team now models total hyperscaler capex of ~US\$1.4tr/US\$1.6tr in 2027/2028



Source: Company data, Morgan Stanley Research estimates, Note: AMZN represents total company capex, SPCX represents terrestrial compute capex. MSFT capex estimates are calendar year.

Exhibit 44: Our US team's bottom-up work shows how total hyperscaler compute capacity is set to reach ~140 GW by 2028 – up from ~30 GW in 2025



Source: Company data, Morgan Stanley Research estimates. Note: SPCX represents terrestrial compute capacity. Note: GOOGL capacity represents additions for both internal GOOGL and Google Cloud, and we expect ~1/2 of capacity additions to go towards each. MSFT capacity estimates represent fiscal year estimates. META capacity estimates include both internally owned and externally leased capacity.

3) Scale-Out Architecture (Rack-to-Rack)

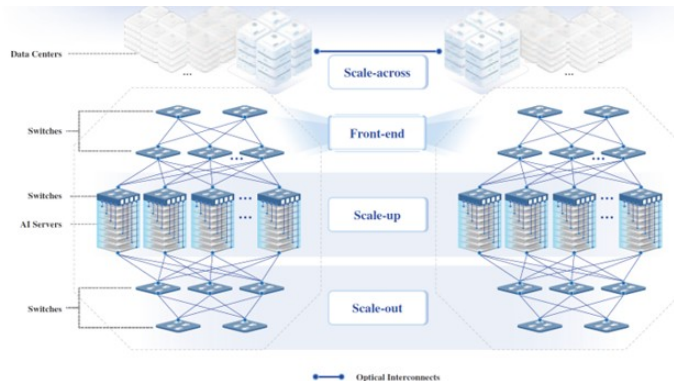
Scale-out networks are currently deployed in AI data centers to connect racks within a data center, enabling efficient scaling for training across hundreds of thousands of GPUs. [We expect widespread scale-out adoption to begin in 2026.](#) Scale-out drives transceiver demand through i) network architecture expansion requiring extensive rack-to-rack connectivity, ii) growing data center clusters with bigger Gen AI clusters, iii) increasing bandwidth requirements pushing the market toward 800G and 1.6T products, and (iv) multiple connection points at various levels of data center infrastructure.

4) Scale-Up Architecture (In-Rack GPU-to-GPU)

Scale-up networking refers to high-speed, low-latency interconnects that link GPUs or other accelerators within a single server or tightly coupled rack. Currently, this traffic is primarily handled by passive copper, but the transition to optical solutions for scale-up applications creates new opportunities for transceiver suppliers. [We expect scale-up to be widely applied from 2027.](#) Network saturation is a critical issue, with up to 33% of GPU time wasted waiting for network availability, translating to US\$10,000+ per GPU per year in idle costs and driving urgent demand for higher-capacity transceivers.

5) Scale-Across Architecture (Data Center-to-Data Center)

Scale-across refers to connecting geographically distributed data centers to operate as a single logical system. The amount of data leaving a data center is growing from approximately 5% of traffic to closer to 20%, creating substantial upward pressure on demand estimates. Scale-across networks depend heavily on optical fiber and optical transceivers for efficient cross-data center collaboration, particularly for long-distance links. Scale-across networks require substantially higher bandwidth than traditional WAN connections – scale-out networks have approximately 10x more capacity than WAN, and scale-up networks have another 10x beyond that.

Exhibit 45: Scale-across, Front-end, Scale-up, and Scale out Architectures

Source: Company data.

Scale-out networks

Scale-out networks are currently deployed in AI data centers to connect racks within a data center, enabling efficient scaling for training across hundreds of thousands of GPUs. The technology addresses connectivity requirements between racks in a data center and is distinct from scale-up (within a rack) and scale-across (between data centers).

Scale-out has historically been the traditional method for data center expansion – they have relied on scaling out by adding more servers to meet growing demand for compute, memory, and storage. However, modern AI requirements are driving demand for more advanced scale-out solutions with higher bandwidth and specialized networking capabilities. According to Lumentum's comments during its results briefing in December 2025, scale-out opportunities will emerge "first" before scale-up applications, with optical technology "connecting from a rack to a switch bank." We believe widespread adoption of advanced AI scale-out architectures has kicked off in 2026, while scale-up is more likely to start from 2027.

We believe broader adoption of scale-out technology could raise transceiver demand for four reasons:

1) Network architecture expansion: Scale-out networks require extensive rack-to-rack connectivity within AI data centers. As hyperscale customers expand GPU clusters with more connected AI nodes and server racks, each AI node must be connected to others in the cluster by fiber. This directly creates more demand for optical transceivers. The scale-out network is a multi-tier infrastructure that connects hundreds or thousands of GPU servers or racks, requiring transceivers at each connection point.

2) Growing data center clusters: The primary driver is the deployment of larger AI data centers with bigger Gen AI clusters and more GPUs. As hyperscalers require more optics for rack-to-rack connectivity and switch ASIC bandwidth increases, demand rises for higher-performance transceiver components and larger volumes across different reach categories – whether short reach (a few meters to 100 meters) or medium-reach (up to a few kilometers).

3) Increasing bandwidth requirements: Scale-out technology drives demand for higher-speed transceivers as bandwidth needs escalate. The market is moving toward 800G and 1.6T products to handle these requirements. Even as the industry shifts to higher speeds, the number of transceivers continues to increase because the overall market is expanding significantly.

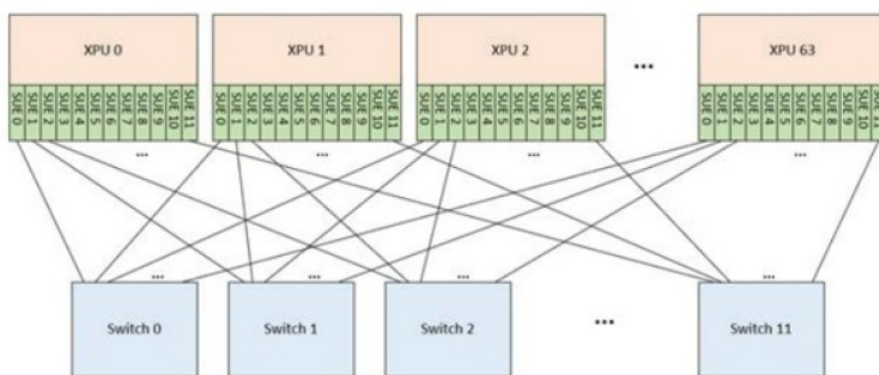
4) Multiple connection points: Scale-out networks create demand across several layers of data center infrastructure, including top-of-rack-switch connections, switch-rack-to-switch-rack interconnects, GPU-rack-to-switch-rack connectivity, and data center interconnects for distributed networks.

Scale-up networks

Scale-up networking refers to high-speed, low-latency interconnects that link GPUs or other accelerators within a single server or tightly coupled rack. This is distinct from scale-out networking, which connects different clusters or data centers. Scale-up networks allow accelerators to operate as a unified computational resource, sharing memory and executing workloads as though they were part of a single, larger system. Key characteristics of scale-up networks include:

- Short-reach connections: Typically, less than 1-2 meters, covering chip-to-chip, board-to-board, backplane, or direct attach cable (DAC) connections.
- Ultra-low latency: Nanosecond-level latency, critical for tightly coupled workloads.
- Extremely high bandwidth: Up to TBps per accelerator.
- Current scale: Today's scale-up architecture typically connects 64-72 accelerators within a rack, although this is likely to grow to 100+ accelerators over time.

Exhibit 46: Example Single Hop Switched SUE Deployment



Source: Scale Up Ethernet Framework White Paper.

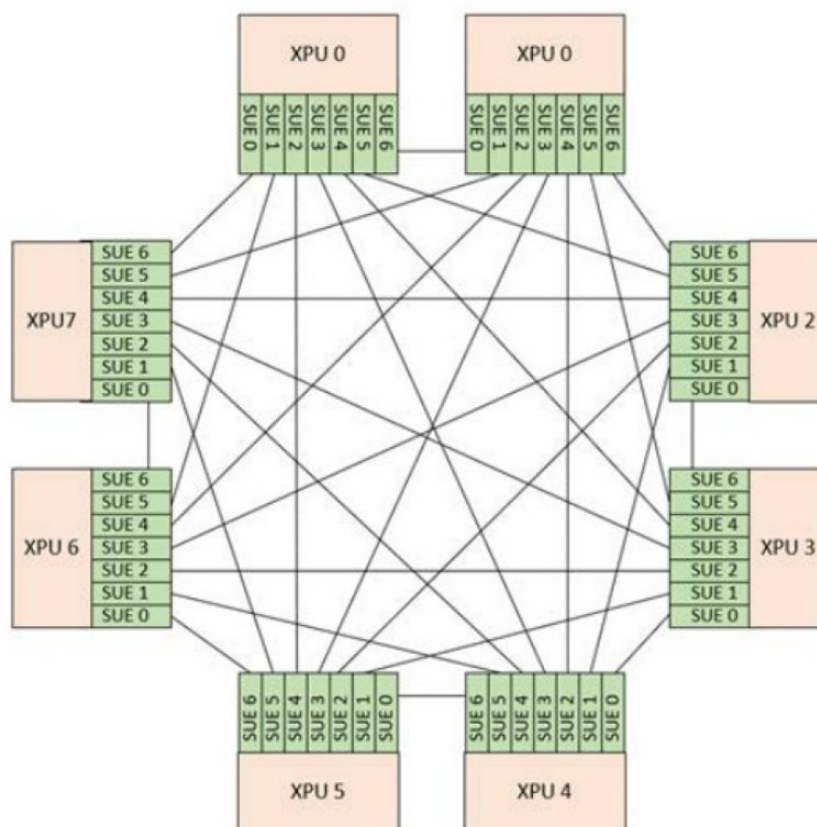
We believe the scale-up technology in data centers could also boost transceiver demand for the following reasons:

1) Increased bandwidth requirements: The scale-up portion of data centers handles 80-90% of traffic, representing XPU-to-XPU communication inside the rack. Currently, this traffic is primarily handled by passive copper. The transition to optical solutions for scale-up applications creates new opportunities for transceiver suppliers.

2) Complex networking and protocol requirements: Scale-up networks involve more complex networking, switching, and routing between GPUs and servers. New protocols are emerging that require interoperability to ensure that different components work together in a rack. This standardization process creates additional opportunities for transceiver deployment because data centers need to verify that components and switches function properly together.

3) Hardware scalability challenges: Rapid growth in AI software systems can be sustained only by hardware that meets the challenges of increasing speed and bandwidth, lower latency, lower power consumption, lower cost, and the ability to scale to volumes required by data centers globally. Network saturation is a critical issue, with up to 33% of GPU time wasted waiting for network availability, translating to US\$10,000+ per GPU per year in idle costs. This inefficiency drives urgent demand for higher-capacity transceivers to eliminate bottlenecks.

Exhibit 47: Example of Mesh SUE Deployment



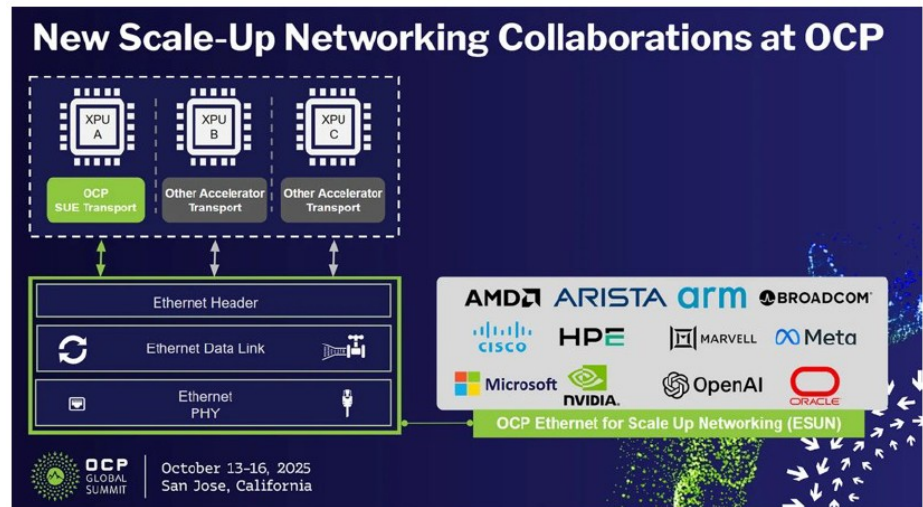
Source: Scale Up Ethernet Framework White Paper.

Several technologies are currently competing to enable scale-up networks, including:

- NVLink: NVIDIA's proprietary solution, currently the dominant technology.
- UALink (Ultra Accelerator Link): An open standard backed by AMD, Astera Labs, Marvell, and hyperscalers.
- Scale-up Ethernet (SUE): Broadcom's AI-optimized Ethernet solution.
- PCIe: Used for custom fabrics and scale-up connections.

As illustrated in the next chart, one of the bigger announcements at Open Compute Project 2025 was ESUN (Ethernet for Scale-Up Networking). The initiative, sponsored by AMD, Arista, ARM, Broadcom, Cisco, HPE, Marvell, Meta, Microsoft, NVIDIA, and Oracle, uses Ethernet as the foundation for an open standard network. Competition among different technology approaches could accelerate industry development. Regardless of which approach becomes the first winner, the transceiver industry is well positioned to benefit from this technology breakthrough.

Exhibit 48: ESUN Announcement Partners



Source: OCP

Scale-across technology

Scale-across refers to connecting geographically distributed data centers so they operate as a single logical system, enabling distributed AI training, inference, storage, and scheduling across multiple sites. This differs from scale-up (within a rack) and scale-out (between racks in a single data center), because scale-across extends computing resources across multiple data centers, facilities, or campuses.

The technology allows hyperscalers to bypass traditional WAN constraints by connecting scale-out networks directly between data centers, effectively making distributed facilities operate as though they were within a single data center. This is particularly important because the amount of data leaving a data center is growing from approximately 5% of traffic to closer to 20%, creating substantial upward pressure on demand estimates.

We believe scale-across technology could also drive transceiver demand for two reasons:

1) Scale-across could alleviate power and physical constraints

By distributing AI training clusters across multiple geographically separated data centers, hyperscalers can access more electrical power while maintaining the processing capability needed for large-scale AI applications. This distributed architecture requires high-speed optical connections between facilities, driving demand for transceivers.

2) Scale-across could meet AI workload requirements

Modern AI inference models are becoming so large that a single data center may not be able to contain them, making it necessary to run inference models across multiple properties. These AI applications demand immense computing power and generate vast data sets that need to be processed and moved rapidly across distributed resources. Scale-across networks depend heavily on optical fiber and optical transceivers for efficient cross-data center collaboration, particularly for long-distance links.

Scale-across architecture requires substantially higher bandwidth than traditional WAN connections. While traditional WANs have limited bandwidth, scale-out networks have approximately 10x more capacity, and scale-up networks have another 10x beyond that. Hyperscalers are essentially trying to "cram scale-out capacity through the WAN," which wasn't built for that amount of capacity. This creates demand for high-speed transceivers, including 800G and 1.6T products.

From a technology perspective, scale-across networks require specialized optical transceivers that can support:

- Long-distance, high-speed connections that traditional copper-based connections cannot support;
- Low latency and high capacity to support real-time AI workloads across distributed sites;
- Advanced features, such as deep buffers, to accommodate link flaps and other issues that occur when scaling across data centers rather than within a single facility.

Silicon photonics (SiPh) is a major driver of rapid 1.6T and 3.2T penetration

Silicon photonics is becoming key to maintaining rapid data growth, supporting AI, and meeting high-performance computing's high-bandwidth needs. SiPh offers lower power consumption (~15% power savings), lower production costs (10–20% cost savings vs. conventional methods), and higher reliability through tighter integration. As the industry migrates from 800G to 1.6T, SiPh transceivers are gaining traction. Google was among the first US CSPs to adopt SiPh optical solutions, and other major CSPs, including Microsoft, AWS, Meta, and Cisco, are increasing adoption of SiPh-based optical transceiver solutions.

We believe rapid growth in SiPh adoption is highly likely, given that GlobalFoundries, an upstream semiconductor vendor, reported that its optical networking serviceable addressable market (SAM) is expected to grow at roughly a 40% CAGR through the end of the decade. GlobalFoundries also reported that silicon photonics revenue doubled in 2025 and is expected to nearly double again in 2026, with a path to a US\$1 billion+ annualized silicon photonics run-rate by the end of 2028.

As an early mover in SiPh technology applications for 1.6T transceivers, we believe InnoLight is highly likely to capture this significant growth opportunity.

Major Technology Developments Changing the Industry Outlook

Co-Packaged Optics (CPO) – Structural Shift, Medium-Term Risk

CPO is an advanced packaging technology that leverages silicon photonics. The core idea is to integrate optical engines directly into the package of a switch chip or other processing chip, enhancing data transmission speeds, shortening transmission distance, reducing power consumption, and minimizing latency.

We believe CPO is unlikely to take share from traditional transceivers in the near term. It could introduce competition only when large-scale application becomes mature, which we believe is more likely in 2027 or later. Combining the current state of CPO development with the expected timing of broad scale-up adoption, we believe meaningful CPO adoption is likely to begin in 2027-2028, with the initial crossover expected at 3.2T and beyond.

Morgan Stanley Research's semiconductor team expects to see more CPO product launches and more concrete development timelines starting in 2H26, including both scale-up and scale-out applications. At GTC 2025, NVIDIA launched its Quantum X800 and Spectrum-X CPO switches. At CES 2026, NVIDIA provided more detail on Spectrum's product plan, which is considered the recommended Ethernet switch product alongside the new GPU Vera Rubin platform. At GTC on March 16, NVIDIA officially launched Spectrum-X CPO switches and announced that CPO technology has moved into full, mass-production, marking a significant shift from pluggable optics to CPO for AI data centers. The company believes the CPO technology is designed to directly address the bandwidth and power consumption challenges in large-scale AI factories.

Our semiconductor team's supply chain checks suggest that currently TSMC plans to ramp-up PIC capacity from 10kwpm in 2Q26 to 15kwpm in 4Q26. They are seeing customers of its Silicon Photonics platform (COUPE) request a very aggressive expansion schedule starting from 1Q27, partially to prepare for scale-up or Optical I/O introduction.

Near-Packaged Optics (NPO) – The "Middle Path" for Transceiver Vendors

NPO transceivers are advanced optical modules in which the optical engine is moved from a traditional pluggable module to a position very close to the switch chip (ASIC) on the same board – typically within 50mm. NPO delivers performance similar to CPO (short electrical path, reduced DSP power, improved signal integrity), while significantly reducing manufacturing complexity.

We view NPO as a critical structural mitigant to CPO disruption risk. If traditional transceiver companies can achieve a breakthrough in NPO technology, it could offset potential disruption from CPO. We note that some overseas clients have placed new NPO orders with leading AI transceiver companies in China – a development that could help NPO mature earlier, alleviating concerns about CPO adoption and potential disruption.

Optical Circuit Switching (OCS) – Emerging Positive Driver

OCS is an emerging networking technology that routes optical signals directly without converting them to electrical signals, eliminating the need for traditional optical-electrical-optical (OEO) conversion. This technology is gaining significant traction in AI data centers due to its ability to deliver ultra-low latency, reduced power consumption, and high reliability.

OCS and optical transceivers form a complementary ecosystem rather than competing technologies – OCS deployment creates additional points where transceivers are needed. Key OCS use cases include spine switch replacement in greenfield deployments, TPU interconnects, and scale-up applications inside the rack. Lumentum expects OCS to reach a US\$1.5 billion+ market by 2029, with initial shipments in 2H25 and ramping across 2026. Key OCS vendors and their solutions include: Lumentum (MEMS-based R300 OCS and R64 OCS), Coherent (Liquid Crystal-based 64x64 and 320x320 platform, shipped to 7 customers as of November 2025), and FSDCS (MEMS-based, launched August 2025).

Market Landscape

Historical Market Share (2016–2022)

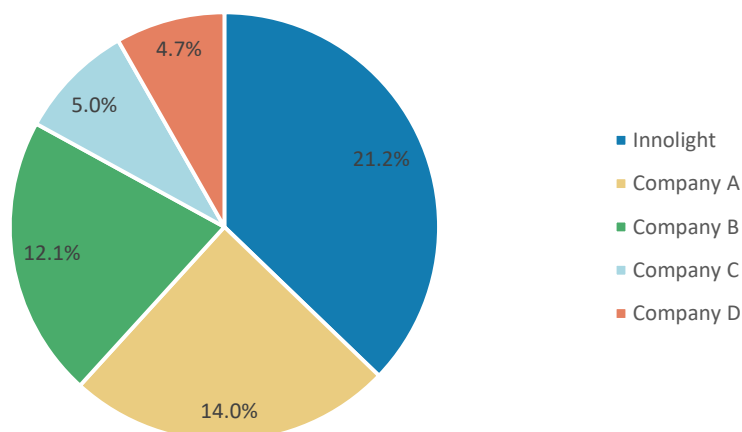
The global transceiver market has undergone significant competitive reshuffling over the past decade. In 2021, the total transceiver revenue market was led by Coherent (formerly Finisar) at 15%, followed by InnoLight at 13%, Accelink at 8%, Lumentum at 8%, Hisense at 7%, Broadcom at 7%, and Cisco at 7%.

In the datacom segment specifically (where AI demand is concentrated), Coherent (Finisar) held 26% share in 2021, followed by InnoLight at 20%, Broadcom at 10%, and Intel at 9%.

InnoLight's global ranking improved dramatically, from No. 7 in 2016 to No. 1–2 (tied with Coherent) in 2022, while Eoptolink rose from outside the top 10 prior to 2018 to No. 7 globally in 2022.

Current Market Dynamics (2024–2025)

In the 800G segment, InnoLight established a first-mover advantage. We estimate the company's market share reached ~30% in 2025, followed by Eoptolink's 22% and Coherent's and Fabrinet's 15%. In the 1.6T segment, we estimate InnoLight had ~37% market share in 2025, followed by Fabrinet's 25% and Eoptolink's 8%.

Exhibit 49: Global ranking of Top 5 interconnect solutions providers (revenues, 2025)

Source: LightCounting, Morgan Stanley Research estimates.

Forward-Looking Market Share Shifts (2026–2028)

In 2026, given that 3.2T transceivers are unlikely to be introduced at scale, we believe more companies could become new vendors of 1.6T products. Eoptolink and Coherent are likely to gain some market share, while Lumentum is more likely to grow broadly with the industry.

We also expect InnoLight to maintain its leading market share position in the AI transceiver industry in absolute dollar terms, even as its percentage share may face some dilution from competitor gains.

Laser Technology Differentiation

As the industry migrates from 800G to 1.6T, the laser type used in transceivers is changing, affecting competitive positioning:

- Silicon photonics (SiPh): InnoLight is best positioned, and this is a key driver of margin improvement.
- Electro-absorption modulated lasers (EML): Our US tech team believes Lumentum and Coherent are well positioned; these lasers offer higher reach but more expensive.
- 200G VCSEL: Our US tech team believes Coherent and Broadcom are well positioned (likely in 2026).

At OFC 2026 (March 2026), several transceiver vendors demonstrated 800G, 1.6T, and even 3.2T products, with 1.6T ramping strongly in 2025 and 3.2T entering customer sampling. The earliest target for 3.2T mass production is 2H27. NVIDIA and Broadcom showed their 1.6T optical engine-embedded ~100T CPO switches (Spectrum-X and Tomahawk 6), which are likely to ship in 2H26.

Supply Chain Constraints and Component Bottlenecks

The 400G/800G transceiver market has faced a supply shortage since 2H24, becoming a bottleneck for the industry. EML supply tightness has been a key constraint, with some transceiver companies starting to use CW (continuous wave) lasers as a substitute for high-end transceivers, especially for 800G/1.6T specifications, given their superior cost-performance and better stability. Currently, both EML and CW lasers face shortages, which is likely to have a negative impact on transceiver companies' shipments.

InP (indium phosphide) substrate supply is another structural constraint. Sumitomo and AXT account for close to 90% of global InP supply, with more than half of global capacity located in China. China's recent export controls on InP substrates have tightened the market, reinforcing the view that InP substrate tightness is structural rather than cyclical.

Geopolitical Risks and Export Control

Geopolitical tensions represent a significant risk factor for the transceiver industry. Key concerns include:

- US export controls on advanced semiconductors and components, which could affect Chinese transceiver manufacturers' access to key inputs.
- China's export controls on gallium and InP substrates, which could benefit upstream Chinese companies relative to Japanese upstream companies.
- Geopolitical tensions that could delay 800G and 1.6T optical engine development for Chinese players with limited overseas revenue exposure.
- China insourcing risk, which could affect pricing and revenue sustainability for companies with exposure to US cloud customers.

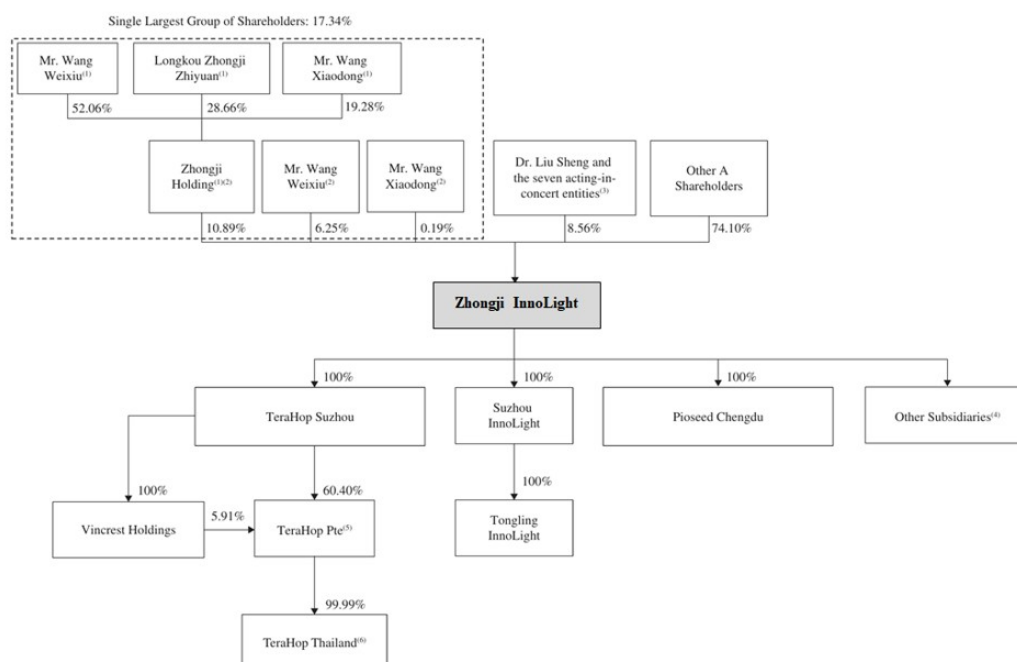
Industry Cyclicity and Historical Patterns

The transceiver industry has historically operated in multi-year capex upcycles. Share price performance for A-share transceiver companies is highly correlated with quarterly revenue trends, with the stocks outperforming during capex upcycles. The current 800G/1.6T cycle, which began in early 2023, is expected to be more sustained than prior cycles because of the structural nature of AI infrastructure investment. Historical cycles (100G, 400G) typically lasted around two years, but the AI-driven cycle is expected to be longer-lasting given the scale of CSP commitments.

Company Background and Strategy

Zhongji InnoLight, a China-based company, is primarily engaged in the research, design, manufacture, and sale of optical transceivers. Its main products include 100G, 200G, 400G, 800G, and 1.6T optical transceivers. The company serves datacom and telecom customers in North America, Europe, and Asia. According to China Insights Consultancy (CIC), InnoLight has been the world's largest optical interconnect solutions provider by revenue for five consecutive years, since 2021.

Exhibit 50: InnoLight: Shareholding structure (as of July 14, 2026)



Source: Company data.

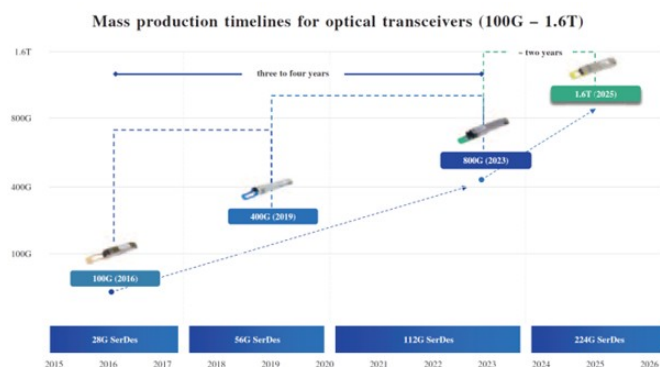
Exhibit 51: InnoLight: Key milestones

Year	Event
2011	Suzhou InnoLight began cooperation with its first tier-one cloud service provider customer and has become the customer's major optical transceiver supplier since then.
2012	The A Shares of Shandong Zhongji Electrical Equipment Co., Ltd. were listed on the ChiNext Market of the Shenzhen Stock Exchange (stock code: 300308).
2013	Suzhou InnoLight's accumulative shipment volume of optical transceivers reached one million.
2014	Google Capital invested in Suzhou InnoLight, marking its first investment in China.
2017	The Company was renamed as Zhongji InnoLight Co., Ltd. following completion of the major asset restructuring of Suzhou InnoLight.
2018	(1) The first 400G optical transceivers in the industry were launched. (2) Global footprint was expanded through the establishment of a Singapore subsidiary. (3) Accumulative shipment volume of optical transceivers reached 10 million.
2020	The first 800G optical transceivers in the industry were launched.
2021	The Company became the world's largest optical interconnect solutions provider by revenue, according to CIC.
2022	Global presence was enhanced with the commencement of operations at the Thailand factory.
2023	The first 1.6T optical transceivers in the industry were launched.
2024	The Company had been the world's largest optical interconnect solutions provider by revenue for four consecutive years, with a market share of 20.6%, according to CIC.
2025	Annualized production capacity exceeded 28 million units.

Source: Company data.

Key products

InnoLight's main products include 100G, 200G, 400G, 800G and 1.6T optical transceivers, which are widely used in data centers, cloud networks, 5G wireless networks and telecom networks. Most of the company's revenue comes from high-end optical transceivers, with overseas revenue accounting for a respective 85% / 87% / 91% of total revenue in 2023 / 2024 / 2025. Client concentration is relatively high, with the top five clients contributing 75%, 77%, and 76% of total revenue for those years.

Exhibit 52: InnoLight: Mass production timeline for optical transceivers (100G-1.6T)

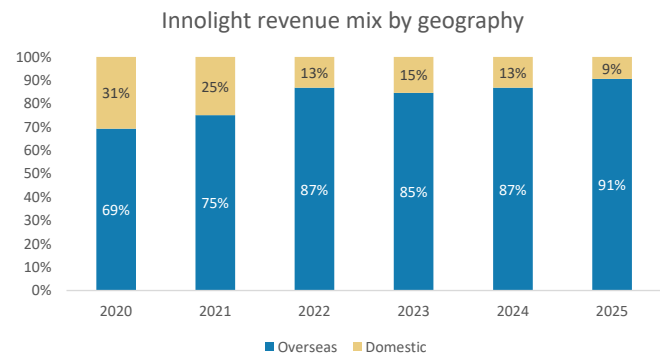
Source: Lightcounting, China Insights Consultancy Ltd.

Exhibit 53: InnoLight: Key product introduction and application scenarios

Product series	Product appearance	Features	Application scenarios
1.6T OSFP		A comprehensive portfolio of 1.6T OSFP 224 optical transceivers, and have taken the industry lead by shipping 1.6T DR8 OSFP224. This series features an 8x200G optical interface design.	Primarily used for 1.6T Ethernet, data centers, and cloud networks.
800G OSFP		A comprehensive portfolio of 800G OSFP 112 optical transceivers, supporting link distance from 100m, 500m, 2km to 10km with VCSEL, EML and SiPh technologies. The company also provide LRO and LPO products under this product series.	Primarily used for 800G Ethernet, data centers, and cloud networks.
800G QSFP-DD		A comprehensive portfolio of 800G QSFP112-DD optical transceivers, supporting link distance from 100m, 500m, 2km to 10km with VCSEL, EML and SiPh technologies.	Primarily used for 800G Ethernet, data centers, and cloud networks.
Coherent 800G ZR OSFP		The company offers 800G ZR OSFP optical transceivers, based on standardized coherent technologies, enabling ultra-high-speed, long-distance data transmission while delivering a strong balance of capacity, performance and interconnectivity, and well-suited for high-bandwidth data center interconnects.	Primarily used for data center interconnects.
400G OSFP		A comprehensive portfolio of 400G OSFP optical transceivers, supporting link distance from 100m, 500m, 2km to 10km with VCSEL, EML and SiPh technologies.	Primarily used for 400G Ethernet, data centers, and cloud networks.
400G QSFP		A comprehensive portfolio of 400G QSFP based optical transceivers, supporting link distance from 100m, 500m, 2km to 10km with VCSEL, EML and SiPh technologies.	Primarily used for 400G Ethernet, data centers, and cloud networks.
Coherent 400G ZR+ QSFP-DD		The company offers 400G ZR+ QSFP-DD optical transceivers, based on standardized coherent technologies, enabling high-capacity, long-distance transmission while balancing power consumption, size and interoperability, and well-suited for data center interconnects.	Primarily used for data center interconnects.
100G QSFP28 Single Lambda		This product series is characterized by a compact form factor, low power consumption, and high data transmission rates.	Primarily used for 100G Ethernet.
100G QSFP28		With low power consumption, compact size, and high data transmission rates, this series enables data centers to increase capacity, improve port density, and reduce overall power consumption.	Primarily used for 100G data center internal networks, data center interconnects, and metropolitan area networks, and can also be applied in 5G wireless networks.

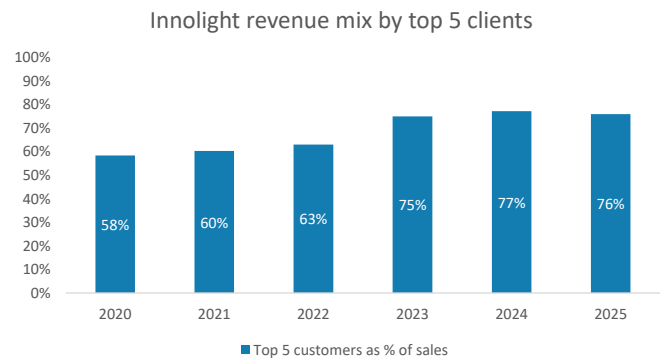
Source: Company data.

Exhibit 54: Revenue mix, by geography



Source: Company data, Morgan Stanley Research.

Exhibit 55: Top five customers' revenue contribution



Source: Company data, Morgan Stanley Research.

Key production bases and global network

InnoLight has established a global footprint supported by five production bases, four R&D centers, and multiple branch offices strategically located around the world. For overseas operations, InnoLight has offices in Singapore and the U.S., primarily for R&D, sales and services, as well as production facilities in Thailand. The overall footprint reflects a China-centric production base with complementary global presence, supporting the company's ability to serve global data center customers.

Exhibit 56: Key production base and offices

Innolight: Key Global Sites



Source: Company data.

Experienced management team

InnoLight appears to have an experienced and seasoned management and board team, combining founder-led industry expertise with functional depth across operations, finance, human resources and corporate governance. The company is led by its founder, Mr. Liu, who has long-standing experience in optical communications R&D and product commercialization. The broader executive team includes senior managers with backgrounds in finance, manufacturing execution and organizational development. At the board level, independent directors with experience in academia, accounting, capital markets and global industrial operations add governance and strategic perspective. Overall, the team profile suggests a balanced mix of technical knowledge, operating experience and oversight capabilities, which should support the company's continued execution as it scales.

Exhibit 57: Summary of key management information

Chinese Name	English Name	Age	Position	Time of joining group	Date of appointment as Director or senior management	Roles and responsibilities	Experience Summary
刘圣	Dr. Liu Sheng	[54]	Chairman of the Board, executive Director and president	July 3, 2017	September 6, 2017	Responsible for the overall strategic planning, business development and operational management of the Group	Founder of Suzhou InnoLight; former R&D engineer at Lucent Technology, China R&D head at Pine Photonics and senior R&D manager at Opnext.
王晓东	Mr. Wang Xiaodong	[49]	Executive Director and executive vice president	January 1, 2015	September 6, 2017	Assisting the president on management and business development of the Group	Former assistant to the chairman of the Board and later served as the chief financial officer of the company from May 2017 to March 2018. He was appointed as a Director in September 2017 and was re-designated as an executive Director in January 2026.
王晓丽	Ms. Wang Xiaoli	[46]	Executive Director, vice president and chief financial officer	July 3, 2017	August 8, 2023	Responsible for the overall financial strategy and management of the Group	Former PwC audit senior manager; CPA and senior CIMA member; oversees finance, accounting, and internal controls.
陈彩云	Ms. Chen Caiyun	[44]	Non-executive Director	July 3, 2017	September 17, 2025	Providing advice on the operation and management of the Group	Former marketing manager in the software industry with extensive experience in corporate management and human resources administration.
战淑萍	Ms. Zhan Shuping	[69]	Independent non-executive Director	September 27, 2021	September 27, 2021	Supervising and providing independent opinions and judgment to the Board	Veteran CPA; former investment banker and listed-company CFO/executive; extensive independent director experience.
成波	Dr. Cheng Bo	[64]	Independent non-executive Director	January 13, 2023	January 13, 2023	Supervising and providing independent opinions and judgment to the Board	Professor at Tsinghua University; Deputy director of State Key Laboratory of Automotive Safety and Energy from 2006 to March 2023 and has been the dean of Suzhou Automotive Research Institute of Tsinghua University since 2011.
屈文洲	Dr. Qu Wenzhou	[53]	Independent non-executive Director	August 8, 2023	August 8, 2023	Supervising and providing independent opinions and judgment to the Board	Professor at Xiamen University, Director of the China Capital Market Research Center, and Director of the MBA Center in School of Management; CPA, CFA with extensive experience in capital markets and corporate governance.
黄国滨	Mr. Huang Guobin	[56]	Independent non-executive Director	June [●], 2026	June [●], 2026	Supervising and providing independent opinions and judgment to the Board	Extensive experience in global investment banking and capital markets, having held senior roles at J.P. Morgan, Goldman Sachs and CICC, and currently serves as chairman of PEC International Group Limited and NEXX Global, and as an independent non-executive director or director of several Hong Kong- and A-share listed companies.

Source: Company data.

Business model

InnoLight mainly engages in assembly within the transceiver supply chain, serving downstream datacom and telecom customers.

Procurement model

The company's key raw materials for optical transceiver production include electrical and optical chips, passive optical components, PCBs and mechanical parts. To ensure product quality, the company has implemented stringent supplier selection and procurement controls covering supplier screening and ongoing management, procurement planning, order tracking, incoming inspection and warehousing. A Supplier Quality Management Operation Specification is in place, under which suppliers are evaluated on a monthly, quarterly and annual basis across metrics such as incoming pass rates, defect levels, and responsiveness to quality issues.

Production model

The company's manufacturing activities are largely order-driven, with the capability to customize products to meet specific customer design and technical requirements. The company has established global production bases and operates a multidisciplinary production team covering process, industrial, electronics, software, mechanical and testing engineering. A comprehensive Product Manufacturing Plan (PMP) and Manufacturing Operation Instructions (MOI) have been implemented. Manufacturing is conducted in a Class 10,000 cleanroom with automated production lines, supported by a manufacturing execution system (MES) to ensure process control, traceability, quality assurance and automated test-data management. These systems support consistent product quality and production efficiency.

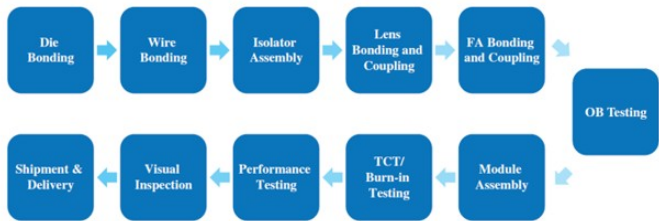
Key processes include production planning, execution and adjustment of production schedules, as well as materials tracking and management. The manufacturing process for optical transceivers primarily consists of several key stages. It begins with attaching optical and electrical dies to the printed circuit board assembly (PCBA), followed by wire bonding to establish electrical links between the dies and the PCBA. This is followed by optical coupling and bonding processes, during which alignment angles and positions are established in connection with optoelectronic conversion, and the fiber array (FA) and lens assemblies are automatically aligned and bonded. Finally, the modules undergo final assembly and performance testing of key parameters to ensure compliance with customer specifications. In addition, during production, the products are subject to temperature cycling testing (TCT) /burn-in testing at the chip level and the module level.

Sales model

High-speed optical modules are core components of downstream products, leading customers to prefer direct and stable cooperation with suppliers. InnoLight mainly adopts a direct sales model, with a small proportion allocated to distribution channels. In 2023, 2024, and 2025, sales generated from direct sale customers amounted to Rmb10,418.8 million, Rmb23,500.9 million, and Rmb37,739.0 million, accounting for 97.2%, 98.5%, and 98.7% of revenue, respectively.

For distribution channels, the company uses distributors primarily for customers with non-recurring or small-volume orders. Distributors operate on a buyer/seller basis and do not act on behalf of the company in dealings with their own customers. All distributors were independent third parties, with no employment, financing or family relationships with the company.

Exhibit 58: Production process



Source: Company data..

R&D model

The company has established dedicated departments for optoelectronic technology, product development, project operations, and comprehensive management, forming a comprehensive technology and product platform with strong in house innovation capabilities. Through talent incentives, training programs, and continuous R&D investment, the company develops new processes and products, improves quality, reduces costs, and enhances profitability. Multiple core technologies support its competitive positioning, supported by robust intellectual property protection measures.

InnoLight's R&D process is designed to drive innovation and ensure the successful development and launch of new products. It begins with comprehensive market research to identify customer needs and industry trends. Based on these insights, the company establishes strategic development routes to determine the direction of product innovation. Next, proposed products undergo a rigorous approval process. Once approved, the product design phase commences, followed by the creation of prototypes and samples for initial evaluation. After refining the design, the company conducts mass production trials to validate manufacturing processes and product performance. Upon successful completion of these trials, products proceed to full-scale mass production.

Exhibit 59: R&D process



Source: Company data.

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
5. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley Asia Limited (which accepts the responsibility for its contents) and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research), and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Andy Meng, CFA.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of July 31, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: AAC Technologies Holdings, Accelink Technologies Co. Ltd., Acer Inc., AirTAC International, Auras Technology Co Ltd, Catcher Technology, Chenbro, Coherent Corp, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Fositek Corp, Genius Electronic Optical Co. Ltd., Gold Circuit Electronics Ltd., Hiwin Technologies Corp., Innolux, Inspur Electronic Information, LandMark Optoelectronics Corporation, LianChuang Electronic Technology Co Ltd, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Lumentum Holdings Inc, Nan Ya PCB, Quanta Computer Inc., Sunny Optical, Suzhou TFC Optical Communication Co Ltd., Unimicron, Visual Photonics Epitaxy Co Ltd, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhejiang Crystal-Optech Co Ltd, Zhen Ding, **Zhongji Innolight Co Ltd**, ZTE Corporation.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Lumentum Holdings Inc, Unimicron, Wistron Corporation, Wiwynn Corp, Zhen Ding, **Zhongji Innolight Co Ltd**.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Coherent Corp, Lenovo, Lumentum Holdings Inc, Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Zhen Ding, **Zhongji Innolight Co Ltd**.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Coherent Corp, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lumentum Holdings Inc, Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, **Zhongji Innolight Co Ltd**.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Coherent Corp, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Lumentum Holdings Inc, Quanta Computer Inc., Xiaomi Corp, Yageo Corp.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Coherent Corp, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lumentum Holdings Inc, Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, **Zhongji Innolight Co Ltd**.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Coherent Corp, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Lumentum Holdings Inc, Quanta Computer Inc., Xiaomi Corp, Yageo Corp., Zhen Ding.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not

linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of August 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1555	42%	450	48%	29%	767	44%
Equal-weight/Hold	1577	43%	395	42%	25%	770	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	559	15%	93	10%	17%	216	12%
Total	3,694		939			1754	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group

Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 14-9169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of AAC Technologies Holdings, BYD Electronics, FIT Hon Teng Ltd, Hon Hai Precision, Lenovo, Lens Technology, Luxshare Precision Industry Co., Ltd., Sunny Optical, Xiaomi Corp, Yangtze Optical Fibre and Cable JSC Ltd, **Zhongji Innolight Co Ltd**, ZTE Corporation listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: <http://www.hkex.com.hk>.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products. Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (09/04/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK\$42.96
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb167.70
BYD Electronics (0285.HK)	O (04/28/2023)	HK\$25.16
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.75
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb16.31
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb386.00

Genius Electronic Optical Co. Ltd. (3406.TW)	E (04/23/2025)	NT\$1,055.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb23.41
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.49
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb34.87
Largan Precision (3008.TW)	O (07/10/2026)	NT\$7,400.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb6.87
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb12.63
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK\$6.63
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK\$59.00
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb54.30
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb7.51
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK\$5.85
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb62.17
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK\$42.10
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb56.78
Sunny Optical (2382.HK)	E (03/23/2026)	HK\$70.50
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb248.70
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb18.13
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK\$28.44
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb378.80
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK\$159.40
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb107.55
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.89
Zhongji Innolight Co Ltd (300308.SZ)	O (09/06/2026)	Rmb814.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK\$23.40
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb33.07
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT\$2,100.00
Advantech (2395.TW)	O (01/20/2021)	NT\$711.00
AirTAC International (1590.TW)	O (04/16/2025)	NT\$1,395.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT\$3,570.00
AU Optronics (2409.TW)	E (02/10/2026)	NT\$26.80
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT\$1,485.00
Bizlink (3665.TW)	O (03/10/2025)	NT\$2,145.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.61
Chenbro (8210.TW)	O (07/23/2025)	NT\$986.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT\$2,030.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT\$1,825.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT\$148.50
Ennostar Inc (3714.TW)	U (09/23/2022)	NT\$66.80
Fositek Corp (6805.TW)	O (06/25/2025)	NT\$2,200.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT\$363.00
Innolux (3481.TW)	E (04/07/2025)	NT\$48.50
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT\$13,860.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb35.14
Lite-On Technology (2301.TW)	E (01/15/2025)	NT\$308.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.80
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.59
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb187.66
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT\$32.70
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT\$1,025.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT\$209.50
Compal Electronics (2324.TW)	U (04/23/2025)	NT\$41.60
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK\$5.31

Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb63.69
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT\$361.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT\$1,045.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT\$256.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb77.72
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT\$3,170.00
Lenovo (0992.HK)	O (07/10/2026)	HK\$32.40
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT\$1,685.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT\$1,055.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT\$95.40
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT\$345.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb137.50
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb331.80
Unimicron (3037.TW)	O (02/23/2026)	NT\$902.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT\$528.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT\$198.00
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT\$2,565.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT\$562.00
Zhen Ding (4958.TW)	++	NT\$463.00

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.